
Diffusion on Hyperbolic Space with a Curvature Determined Confinement

Anonymous Author(s)

Affiliation

Address

email

Abstract

Score based generative models need a forward stochastic process whose law converges to a reference distribution that can be sampled. Hyperbolic space $\mathbb{H}^n$ is a natural state space for hierarchical data, but Brownian motion on it is transient and has no invariant probability law. We analyse the confined diffusion $dX_t = -\frac{c}{2}\nabla r^2 dt + \sqrt{2} dB_t$, whose invariant law $\mu_c \propto e^{-cr^2/2} d\text{vol}$ exists for every $c > 0$, and trace its spectral structure into the design, training and validation of a generative model. The curvature threshold $c = n - 1$ separates two regimes of the spectral gap $\lambda_1(c)$: above it $c - (n - 1) \leq \lambda_1(c) \leq \text{Var}_{\mu_c}(r)^{-1} - c$, so the gap grows linearly in c , while below it $\lambda_1(c)$ collapses faster than any power of c , at least as fast as $\sqrt{c} e^{-1/(2c)}$ on $\mathbb{H}^2$, although it stays positive. A commutation identity shows that the gap is attained in the first spherical harmonic sector for every c and that radial observables relax exactly c faster than angular ones; this identifies the slow mode a sampler must reverse and makes the angular order parameter an exact probe of $\lambda_1(c)$, and simulated relaxation rates agree with numerically computed gaps to within four percent in dimensions two to four. The score is anchored at the stationary drift and read in an orthonormal frame, since extrinsic network outputs are amplified by $\Theta(e^r)$ and cause divergence in our ablations. On hierarchical data, moving the confinement across the threshold degrades generation by a factor of thirteen with architecture, budget and estimator fixed, and a statistic computed from the forward process alone predicts this failure before training; above the threshold the intrinsic model matches a wrapped normal baseline trained with exact score targets.

1 Introduction

Denoising diffusion and score based models generate data by reversing a stochastic process: the forward process carries the data law towards a reference law, and a learned reverse process transports it back [1, 2]. On $\mathbb{R}^n$ the standard choice is a variance preserving Ornstein–Uhlenbeck (OU) process with a Gaussian reference law [2]. On compact manifolds, Brownian motion already has the normalised volume as a stationary law, which is used for spherical and rotational data [3], for torsion angles on tori [5] and, combined with Euclidean translational noising, for rigid body and protein geometry [6, 7]; E(3)-equivariant Euclidean diffusion is also used for three-dimensional molecular coordinates [8]. Discrete state spaces use Markov corruption processes [9, 10].

The compact construction does not extend to hyperbolic space $\mathbb{H}^n$, a natural state space for hierarchical data: trees admit embeddings in $\mathbb{H}^2$ with arbitrarily low distortion [13], while hyperbolic embeddings are effective for learned hierarchical representations [12]. The space $\mathbb{H}^n$ has infinite

35 volume, Brownian motion on it is transient [27], and the bottom of the spectrum of $-\Delta$ is $(n-1)^2/4$
 36 [18, 26]; undrifted Brownian motion therefore has no stationary probability law. Alternatives are the
 37 wrapped normal, a Euclidean Gaussian pushed through the exponential map [11], and Riemannian
 38 diffusion models with a drift towards a prescribed prior [3, 4]. We study radial quadratic confinement,
 39 $W_c = \frac{c}{2}r^2$ with generator $\mathcal{L}_c = \Delta - \nabla W_c \cdot \nabla$ and invariant law $\mu_c \propto e^{-cr^2/2} d\text{vol}$, and ask when
 40 the stationary law exists, how quickly the process mixes and how those rates affect a generative
 41 model.

42 The paper follows the confined diffusion from its stochastic analysis to a sampler, organised around
 43 two questions: how the invariant law and the spectral gap of the forward process guide the choice of
 44 c and supply a validation statistic that can be evaluated before any network is trained, and how the
 45 structure of the stationary dynamics can be built into the reverse process so that the sampler stays
 46 stable when its inputs leave the training support. Our contributions are as follows.

- 47 • **Spectral estimates on both sides of the threshold** (Section 3). The law μ_c exists and is the
 48 unique invariant probability measure for every $c > 0$; above the threshold $c = n - 1$ the Bakry–
 49 Émery bound gives $\lambda_1(c) \geq c - (n - 1)$ and a Rayleigh quotient an upper bound of the same
 50 order (Theorems 3.1, 3.2). A commutation identity (Proposition 3.3) shows that the nonzero radial
 51 spectrum of $-\mathcal{L}_c$ is the first spherical harmonic spectrum shifted by exactly c : the gap is attained
 52 in the first harmonic sector for every c , radial observables relax c faster than angular ones, and
 53 $\lambda_1(c) \leq \text{Var}_{\mu_c}(r)^{-1} - c$ (Corollary 3.4); a formal expansion gives $\lambda_1(c) = c - \frac{2}{3}(n-1) + O(c^{-1})$
 54 (Appendix B.7). Below the threshold we prove an exponential collapse with a numerically sharp
 55 exponent (Theorem 3.5) while the gap stays positive (Remark 3.6): the threshold is a crossover in
 56 the mixing rate, not a boundary of existence.
- 57 • **A stable score model** (Section 4). The score is the stationary drift plus a learned residual in
 58 an orthonormal frame, so a zero residual recovers the ergodic stationary process exactly, and
 59 the frame avoids the $\Theta(e^r)$ amplification of extrinsic outputs under the Minkowski projection
 60 (Lemma 4.1), which otherwise creates a feedback loop once a sample leaves the training support.
- 61 • **Experiments that validate the theory and resolve the two regimes** (Section 5). Simulated
 62 relaxation rates on $\mathbb{H}^2$, $\mathbb{H}^3$ and $\mathbb{H}^4$ agree with finite element spectral gaps to within four percent,
 63 and radial observables relax faster than angular ones by the amount c that Proposition 3.3 predicts.
 64 On hierarchical data, changing only c from 2 to 0.25 degrades generation by a factor of thirteen,
 65 and the failure is visible in the forward process before training. Section 2 and Appendix A place
 66 $\mathbb{H}^n$ among the geometries in use: finite volume, not compactness, decides whether Brownian
 67 motion needs a drift.

68 **Related work.** Riemannian diffusion models [3, 4] develop forward and reverse diffusions and
 69 trainable objectives on manifolds, including synthetic non-compact hyperbolic examples with a drift
 70 towards a chosen prior. Our question is more specific: we analyse radial quadratic confinement on
 71 $\mathbb{H}^n$, identify its curvature dependent spectral regimes and trace them into score parameterisation and
 72 generative failure. Time reversal is classical [14, 15]; score matching and its denoising form are due to
 73 Hyvärinen [16] and Vincent [17]. Wrapped normal models [11] and hyperbolic representation learning
 74 [12, 13] motivate the geometry; the spectral analysis uses Bakry–Émery theory [29], Lyapunov type
 75 Poincaré criteria [19] and the polar-decomposition isospectrality of A^*A and AA^* [21], with Deift’s
 76 commutation method providing a related spectral precedent [20]. We do not claim that radial quadratic
 77 confinement is the optimal generator; our aim is to characterise its geometric cost.

78 2 Brownian motion, invariant measures and the need for a drift

79 Let (M, g) be a smooth connected Riemannian manifold without boundary, with volume vol_g and
 80 Laplace–Beltrami operator Δ (unit Laplacian convention), so that Brownian motion has generator Δ
 81 and, when conservative, semigroup $P_t = e^{t\Delta}$ [27]. A probability measure is invariant if P_t preserves
 82 it; vol_g is always infinitesimally invariant (Proposition A.1), so the question is whether it can be
 83 normalised.

84 **Theorem 2.1** (A dichotomy between finite and infinite volume). *Let (M, g) be connected and without*
 85 *boundary, and assume Brownian motion is conservative.*

86 (i) *If $\text{vol}_g(M) < \infty$, the normalised measure $\bar{\mu} = \text{vol}_g / \text{vol}_g(M)$ is the unique invariant probabil-*
 87 *ity measure of (X_t) .*

88 (ii) *If, in addition, M is compact, then $-\Delta$ has discrete spectrum $0 = \lambda_0 < \lambda_1 \leq \dots$ and*
 89 $\|P_t f - \bar{\mu}(f)\|_{L^2(\bar{\mu})} \leq e^{-\lambda_1 t} \|f - \bar{\mu}(f)\|_{L^2(\bar{\mu})}$; *the process mixes exponentially at the spectral*
 90 *gap.*

91 (iii) *If $\text{vol}_g(M) = \infty$, there is no invariant probability measure of constant density. Let V satisfy*
 92 $e^{-V} \in L^1(\text{vol}_g)$, *and suppose the diffusion generated by $\mathcal{L}_V = \Delta - \nabla V \cdot \nabla$ is conservative.*
 93 *Then $d\mu_V \propto e^{-V} d\text{vol}_g$ is its unique invariant probability law.*

94 The proof is in Appendix A. The theorem separates two roles often assigned to compactness: finite
 95 volume normalises the volume measure, whereas compactness supplies a discrete spectrum and
 96 a positive gap; a hyperbolic surface of finite area that is not compact has the normalised volume
 97 as stationary law but essential spectrum $[1/4, \infty)$ [26] (Proposition A.2). Because $\mathbb{H}^n$ has infinite
 98 volume, Theorem 2.1(iii) forces a drift. Appendix A records the rest of the classification for the
 99 geometries in common use (symmetry, boundary, discreteness); the remainder of the paper quantifies
 100 what the drift costs on $\mathbb{H}^n$.

101 3 The hyperbolic Gaussian and its spectral gap

102 We work on $\mathbb{H}^n$, $n \geq 2$, in the hyperboloid model $\{x \in \mathbb{R}^{n+1} : \langle x, x \rangle_L = -1, x_{n+1} > 0\}$, where
 103 $\langle x, y \rangle_L = \sum_{i \leq n} x_i y_i - x_{n+1} y_{n+1}$ and the sectional curvature is -1 . We take $o = (0, \dots, 0, 1)$ as
 104 the base point, write $r(x) = d(o, x)$ and, for $c > 0$, set

$$W_c = \frac{c}{2} r^2, \quad \mathcal{L}_c = \Delta - \nabla W_c \cdot \nabla, \quad dX_t = -\nabla W_c(X_t) dt + \sqrt{2} dB_t,$$

105 with unnormalised invariant density e^{-W_c} with respect to $d\text{vol}$. Write $\lambda_1(c)$ for the spectral gap of
 106 $-\mathcal{L}_c$ in $L^2(\mu_c)$, equivalently the optimal Poincaré constant

$$\lambda_1(c) = \inf \left\{ \frac{\int |\nabla f|^2 d\mu_c}{\text{Var}_{\mu_c}(f)} : f \text{ smooth, bounded, nonconstant} \right\}.$$

107 In geodesic polar coordinates about o , $d\mu_c = \rho_c(s) ds d\sigma(\theta)$ with $\rho_c(s) = Z_c^{-1} \sinh^{n-1}(s) e^{-cs^2/2}$
 108 and σ the normalised measure on S^{n-1} .

109 3.1 The confined diffusion exists for every $c > 0$

110 **Theorem 3.1** (A hyperbolic Gaussian whose gap is set by curvature). *On $\mathbb{H}^n$, for $\mathcal{L}_c$ as above:*

111 (i) *for every $c > 0$, $d\mu_c \propto e^{-\frac{c}{2} r^2} d\text{vol}$ is a probability measure and the unique invariant law of the*
 112 *diffusion, which is non-explosive;*

113 (ii) *if $c > n - 1$, then $\text{Ric} + \nabla^2 W_c \geq (c - (n - 1))g$ and $\mathcal{L}_c$ satisfies the Bakry–Émery condition*
 114 $\text{CD}(c - (n - 1), \infty)$; *hence $\lambda_1(c) \geq c - (n - 1)$ and*

$$\|P_t f - \mu_c(f)\|_{L^2(\mu_c)} \leq e^{-(c-(n-1))t} \|f - \mu_c(f)\|_{L^2(\mu_c)}.$$

115 The proof (Appendix B) sets the Gaussian decay of $e^{-cr^2/2}$ against the volume growth $\sinh^{n-1} r$
 116 and uses the Hessian comparison $\nabla^2(\frac{1}{2}r^2) \geq g$. The threshold $c = n - 1$ is the magnitude of the
 117 negative Ricci term, above which the convexity of the potential dominates it; for curvature $-a^2$ the
 118 condition is $c > (n - 1)a^2$, and $a \rightarrow 0$ recovers the Euclidean OU process.

119 3.2 An upper bound of matching order

120 **Theorem 3.2** (An upper bound of matching order). *For every $c > 0$,*

$$\lambda_1(c) \leq \frac{1}{\text{Var}_{\mu_c}(r)}.$$

Table 1: Spectral gap of $-\mathcal{L}_c$ on $\mathbb{H}^2$ against the bounds of this section. Numerical values are the lowest eigenvalues of the sector problems (1) by a linear finite element method (grid refinement changes them by less than 10^{-6} relative); simulated rates are from Figure 1; n/a marks an unreached fitting window or a vacuous bound.

c	0.1	0.25	0.5	1	2	4	8
$\lambda^{(1)}(c) = \lambda_1(c)$, numerical	0.00127	0.0379	0.176	0.565	1.474	3.412	7.375
$\lambda^{(0)}(c) - c$, numerical	0.00127	0.0379	0.176	0.565	1.474	3.412	7.375
simulated rate (Section 5.1)	n/a	0.0369	0.174	0.544	1.464	3.368	n/a
lower bound $c - (n - 1)$	n/a	n/a	n/a	0	1	3	7
upper bound $\text{Var}_{\mu_c}(r)^{-1} - c$ (Cor. 3.4)	0.00150	0.0491	0.239	0.775	2.011	4.612	9.899
envelope $\sqrt{c} e^{-1/(2c)}$ (Thm. 3.5)	0.00213	0.0677	0.260	0.607	n/a	n/a	n/a

Here $c \text{Var}_{\mu_c}(r) \rightarrow v_n$ as $c \rightarrow \infty$, where $v_n = n - 2(\Gamma(\frac{n+1}{2})/\Gamma(\frac{n}{2}))^2 \in (0, 1]$ is the variance of a χ_n random variable. Consequently there is $c_0(n)$ with

$$c - (n - 1) \leq \lambda_1(c) \leq \frac{2}{v_n} c \quad \text{for all } c \geq c_0(n).$$

Thus the spectral gap grows linearly in the confinement and Theorem 3.1(ii) is order sharp.

The first inequality tests the Rayleigh quotient with the 1-Lipschitz function $f = r$; for large c the radial law approaches a χ_n law at scale $c^{-1/2}$, so the bound has the Euclidean scaling of the OU gap c .

3.3 The gap sits in the first spherical harmonic sector

Since $\mathcal{L}_c$ commutes with the rotations fixing o , $L^2(\mu_c)$ decomposes into sectors indexed by the degree $\ell \geq 0$ of spherical harmonics on S^{n-1} , and on the sector spanned by $g(r)Y(\theta)$ with Y of degree ℓ the operator $-\mathcal{L}_c$ acts on the radial factor as

$$-\mathcal{L}_c^{(\ell)} = -\partial_s^2 - a(s)\partial_s + \frac{\ell(\ell + n - 2)}{\sinh^2 s}, \quad a(s) := (\log \rho_c)'(s) = (n - 1) \coth s - cs, \quad (1)$$

with Dirichlet form $\mathcal{E}_\ell(g) = \int_0^\infty (g'^2 + \ell(\ell + n - 2) \sinh^{-2}(s) g^2) \rho_c ds$ on $\mathcal{H} := L^2((0, \infty), \rho_c ds)$.

Write $\lambda^{(\ell)}(c)$ for the bottom of the spectrum of $-\mathcal{L}_c^{(\ell)}$, excluding the eigenvalue 0 of the constants when $\ell = 0$. The following identity is the hyperbolic analogue of the OU ladder, where ∂_{x_1} maps the radial eigenfunction $|x|^2 - n/c$ to x_1 and lowers the eigenvalue from $2c$ to c .

Proposition 3.3 (Radial and angular spectra differ by exactly c). *Let $n \geq 2$ and $c > 0$.*

(i) *The nonzero spectrum of $-\mathcal{L}_c^{(0)}$ equals the spectrum of $-\mathcal{L}_c^{(1)}$ shifted by c , with multiplicity. In particular $\lambda^{(0)}(c) = \lambda^{(1)}(c) + c$, and if F is a radial eigenfunction with eigenvalue Λ , then $F'(r)\theta_1$ is an eigenfunction in the degree one sector with eigenvalue $\Lambda - c$.*

(ii) *$\lambda^{(\ell)}(c) \geq \lambda^{(1)}(c)$ for every $\ell \geq 1$.*

(iii) *Consequently $\lambda_1(c) = \lambda^{(1)}(c) = \lambda^{(0)}(c) - c$ for every $c > 0$: the spectral gap is attained in the first spherical harmonic sector, and the slowest radial mode relaxes exactly c faster than the slowest angular mode.*

The proof (Appendix B.3, which also proves the corollary below) writes $-\mathcal{L}_c^{(0)} = D^*D$ and $-\mathcal{L}_c^{(1)} + c = DD^*$ for the closed operator $D = \partial_s$ on $\mathcal{H}$ and uses the polar-decomposition isospectrality of D^*D and DD^* off zero [21]; the shift is c because $-a'(s) = (n - 1) \sinh^{-2} s + c$, the angular potential of the first harmonic plus the confinement. Three consequences matter for the rest of the paper.

Corollary 3.4 (Sharpened upper bound and the slow mode). *For every $n \geq 2$ and $c > 0$,*

$$\lambda_1(c) \leq \frac{1}{\text{Var}_{\mu_c}(r)} - c, \quad \text{and} \quad 1 \leq \liminf_{c \rightarrow \infty} \frac{\lambda_1(c)}{c} \leq \limsup_{c \rightarrow \infty} \frac{\lambda_1(c)}{c} \leq \frac{1}{v_n} - 1.$$

Moreover, if the initial law has a density $p_0/\mu_c \in L^2(\mu_c)$ with a nonzero component along the ground state of the degree one sector, then $\mathbb{E} \theta_1(X_t)$ decays at the asymptotic rate $\lambda_1(c)$, whereas $\mathbb{E} f(X_t) - \mu_c(f)$ decays at least at rate $\lambda_1(c) + c$ for every radial $f \in L^2(\mu_c)$.

The sharpened bound confines $\lambda_1(c)/c$ asymptotically to $[1, v_n^{-1} - 1]$, which is $[1, 1.33]$ on $\mathbb{H}^2$ and shrinks towards $\{1\}$ as $n \rightarrow \infty$. Table 1 reports $\lambda_1(c)$ on $\mathbb{H}^2$ computed from (1) by a linear finite element method (supplied code), and Table 6 in Appendix C does the same on $\mathbb{H}^3$, $\mathbb{H}^4$ and $\mathbb{H}^5$: the identity $\lambda^{(0)} = \lambda^{(1)} + c$ holds to discretisation error throughout, the sharpened bound is tight below the threshold (within 0.5% at $c = 0.25$ for $n = 4, 5$), and the simulated rates of Section 5.1 agree with $\lambda_1(c)$ to within four percent on $\mathbb{H}^2$ and two percent on $\mathbb{H}^3$ and $\mathbb{H}^4$.

A formal expansion at the origin, carried to second order in Appendix B.7 and confirmed there numerically for $n = 2, \dots, 5$, predicts $\lambda_1(c) = c - \frac{2}{3}(n-1) + \frac{4(n-1)(n+2)}{45c} + O(c^{-2})$, so that $\lambda_1(c)/c \rightarrow 1$ and the Bakry–Émery bound of Theorem 3.1(ii) is sharp to leading order; we state this as a prediction and do not use it elsewhere.

3.4 The collapse below the threshold

Theorem 3.5 (Collapse below the threshold). *There are constants $C(n) < \infty$ such that for all $0 < c \leq 1$: (i) on $\mathbb{H}^2$, $\lambda_1(c) \leq C\sqrt{c} e^{-1/(2c)}$; (ii) on $\mathbb{H}^n$, $n \geq 3$, $\lambda_1(c) \leq C(n) e^{-\beta(n)/c}$ with $\beta(n) = \min \left\{ \frac{(n-1)^2}{2}, 2(n-2) \right\}$. In particular $\lambda_1(c) \rightarrow 0$ faster than any power of c as $c \downarrow 0$.*

Slow angular modes drive the bound: below the threshold most equilibrium mass lies in a thin shell near $r_* \approx (n-1)/c$, where a first spherical harmonic cut off near the origin has angular Dirichlet energy weighted by $\sinh^{-2} r \approx 4e^{-2r}$ and the cutoff costs only the exponentially small mass near o ; these two costs give the exponents in $\beta(n)$, the angular one $2(n-2)$ being the smaller for $n \geq 3$, while on $\mathbb{H}^2$ the angular term is subexponential and the radial cost $1/(2c)$ remains. Proposition 3.3 explains why the collapse is angular: $-(\log \rho_c)'' = c + (n-1) \sinh^{-2} s \geq c$, so the radial gap $\lambda^{(0)}(c) = c + \lambda_1(c)$ is at least c for every c while the angular gap is exponentially small. The exponents are consistent with being sharp: least squares fits of numerically computed $\log \lambda_1(c)$ against $1/c$ on $\mathbb{H}^n$, $n = 2, \dots, 5$ (Appendix C), give 0.48, 1.97, 3.94 and 5.98 against the predicted $\frac{1}{2}, 2, 4$ and 6. A matching lower bound is open; by Proposition 3.3(iii) it is equivalent to the second order small c asymptotics of the one dimensional radial gap $\lambda^{(0)}(c) - c$ (Section 6).

Remark 3.6 (Strict positivity for every c). $\lambda_1(c) > 0$ for all $c > 0$: with $V = e^{\alpha r}$ smoothed near o , $\mathcal{L}_c V/V \rightarrow -\infty$, and the Lyapunov criterion of [19] gives a Poincaré inequality (Appendix B). The threshold changes the scale of the gap, not its positivity.

4 From the theorem to a sampler

4.1 Forward and reverse processes

The forward process follows $dX_t = -\nabla W_c dt + \sqrt{2} dB_t$ from the data law until time T , approaching μ_c at a rate set by $\lambda_1(c)$; μ_c has uniform angular component and radial density ρ_c , sampled by inverse transform. The reverse process is the classical time reversal [14, 15], on a manifold as in [3]:

$$dY_t = [\nabla W_c(Y_t) + 2\nabla \log p_{T-t}(Y_t)] dt + \sqrt{2} d\bar{B}_t, \quad (2)$$

with p_t the density with respect to vol. At stationarity $\nabla \log \mu_c = -\nabla W_c$, so the bracket is $-\nabla W_c$ and the stationary process is its own reversal. By Corollary 3.4, the last component of the data law to be forgotten by the forward process is its first angular harmonic, which the reverse process must therefore reconstruct from the least informative endpoint.

4.2 Anchored scores read in a frame

We build this stationary structure into the score parameterisation:

$$s_\theta(x, t) = -\nabla W_c(x) + u_\theta(x, t), \quad \text{and the reverse drift is } -\nabla W_c + 2u_\theta. \quad (3)$$

When $u_\theta \equiv 0$ the sampler is exactly the ergodic stationary diffusion, so training perturbs a dynamics that already confines samples, whereas an unanchored network must cancel the outward $+\nabla W_c$ term in (2) at every radius and a systematic error leaves a net outward drift. The anchor is a structural guarantee outside the training support, where the residual is unreliable: whatever the network outputs there, the confining term remains. The representation of the residual matters at large radius, where the hyperboloid embedding is poorly conditioned.

Lemma 4.1 (Extrinsic outputs are exponentially amplified). *Let $x \in \mathbb{H}^n$ with $r = d(o, x)$ and for $w \in \mathbb{R}^{n+1}$ let $P_x w = w + \langle x, w \rangle_L x$ be the Minkowski projection onto $T_x \mathbb{H}^n$. Then*

$$\sinh r \leq \sup_{|w|_2=1} |P_x w|_g \leq \sqrt{2} \cosh r,$$

the lower value attained at the timelike unit vector. An extrinsic network output of Euclidean size $O(1)$ can therefore produce a tangent vector of size $\Theta(e^r)$. A geodesic step inherits the same amplification (proof in Appendix B).

The amplification creates a feedback loop: once a sample leaves the training support, an inaccurate extrinsic output amplified by $\Theta(e^r)$ moves it to still larger radius. We therefore predict scalar coefficients in an orthonormal frame: on $\mathbb{H}^2$, away from the origin, $u_\theta = a_\theta e_r + b_\theta e_\varphi$ with $e_r = \nabla r$, whose norm is set by the network outputs and does not grow with the embedding coordinates (inputs are bounded functions of (r, φ, t) , Appendix C); for $n \geq 3$ a radial head and a normalised tangential head with a learned amplitude serve the same purpose. In Section 5.2 the unanchored extrinsic, anchored extrinsic and anchored frame variants diverge at rates 28.7%, 27.1% and 0%.

4.3 Training, discretisation and floating point

We train u_θ on simulated forward trajectories using score matching [16] in its denoising form [17], with tangent Gaussian geodesic random walks over short intervals as in Riemannian diffusion models [3]. For our training objective, for a transition from x_k to x_{k+m} over $\tau = mh$ we use the target $\hat{s}(x_{k+m}) = \log_{x_{k+m}}(\Phi_\tau(x_k))/(2\tau)$, with Φ_τ the drift flow, as a first order approximation to the conditional mean. With $m = 1$ the signal to noise ratio is of order $\sqrt{h}$ and the $\sinh^{-1} r$ metric factor suppresses angular displacements at large radius, so we draw m uniformly from $\{1, 2, 4, 8\}$, trading an $O(mh)$ bias for a variance reduction of order $\sqrt{m}$; the wrapped baseline of Section 5.2 uses exact Gaussian conditionals (details in Appendix C).

Both processes are integrated by geodesic Euler–Maruyama with reprojection after each step, $x_{k+1} = \exp_{x_k}(-\nabla W_c(x_k)h + \sqrt{2h}\xi_k)$. Hyperboloid coordinates grow as e^r , so evaluating $\langle x, x \rangle_L = -1$ cancels with error $\varepsilon_{\text{mach}} e^{2r}$ and the usable radius is $r \lesssim \frac{1}{2} \log(2/\varepsilon_{\text{mach}})$, about 8.3 in fp32 and 18.4 in fp64, as measured constraint errors confirm (Table 9, Appendix C); the Poincaré ball has the same sensitivity through its conformal factor. Our implementation therefore uses fp64, reprojects after every step and supplies the radius directly to the frame based features.

5 Experiments

5.1 Relaxation rates across the threshold

On $\mathbb{H}^2$ we integrate the forward SDE in polar coordinates, $dr = (\coth r - cr) dt + \sqrt{2} dW_r$ and $d\varphi = (\sqrt{2}/\sinh r) dW_\varphi$ (double precision, fixed seeds, $N = 20,000$ particles, $T = 6$, $h = 0.005$, $c \in \{0.05, \dots, 4\}$, small initial cloud at radius 3 and angle 0), and fit the decay rate of $R(t) = |\mathbb{E} e^{i\varphi t}|$ on the window $0.05 \leq R \leq 0.7$. By Proposition 3.3(iii) the late time decay rate of $R(t)$ is exactly $\lambda_1(c)$, so the fit estimates the gap rather than a proxy for it: the ratios to $\lambda_1(c)$ from the sector problems are 0.97, 0.99, 0.96, 0.99 and 0.99 at $c = 0.25, 0.5, 1, 2$ and 4 (Table 1; integrator and initialisation checks in Appendix C).

Figure 1 separates the two regimes. Above the threshold the rates lie above the Bakry–Émery line and approach it as c grows; below it they fall to 0.174 at $c = 0.5$ and 0.037 at $c = 0.25$, against 0.26

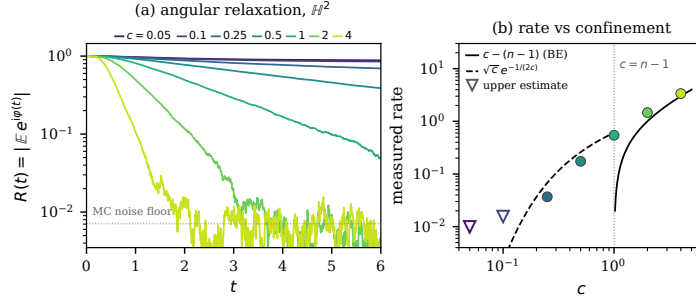


Figure 1: Forward relaxation on $\mathbb{H}^2$. (a) Angular order parameter $R(t)$ for confinements spanning the threshold $c = 1$; the dotted line is the Monte Carlo floor $N^{-1/2}$. (b) Fitted rates against c : above threshold they approach the Bakry–Émery bound $c - (n - 1)$ (solid) from above, with ratios 1.46 at $c = 2$ and 1.12 at $c = 4$; below threshold they collapse along the $\sqrt{c} e^{-1/(2c)}$ envelope of Theorem 3.5 (dashed). Open triangles at $c \leq 0.1$ are transient limited and show only an upper estimate.

and 0.068 for $\sqrt{c} e^{-1/(2c)}$ and 0.176 and 0.038 for the computed gap, and for the two smallest c the trajectories do not mix within the horizon.

Higher dimensions. The same simulation runs on $\mathbb{H}^n$ for any n , since in polar coordinates the forward process is $dr = ((n - 1) \coth r - cr) dt + \sqrt{2} dW_r$ with a Brownian motion on S^{n-1} run at speed $\sinh^{-2} r$. Tables 7 and 8 in Appendix C report rates on $\mathbb{H}^2$, $\mathbb{H}^3$ and $\mathbb{H}^4$ for an angular observable in the first harmonic sector and a radial one, which Proposition 3.3 predicts to relax exactly c faster. With the exact eigenfunctions of the sector operators as observables, which decay as single exponentials for any initial law, the fitted rates agree with $\lambda_1(c)$ and $\lambda_1(c) + c$ to within 2% on $\mathbb{H}^2$, at every step size and for two integrators, and to within 1.5% on $\mathbb{H}^3$ and $\mathbb{H}^4$, and the difference of the two rates divided by c lies between 0.97 and 1.02 in all 19 cases. Rates fitted to the raw observables $\mathbb{E} \theta_1$ and $\mathbb{E} r_t$ agree to within 2% and 6%; the appendix traces the larger radial deviation to fitting a sum of exponentials on a fixed window.

5.2 Generative modelling of hierarchical data

Setup. A two level hierarchy on $\mathbb{H}^2$: four groups at angles $k\pi/2$ with weights $(0.4, 0.3, 0.2, 0.1)$, each split into two leaves at angular offsets ± 0.175 , leaf centres at radius 3.5 with wrapped Gaussian noise of scale 0.15, 8,192 training and 4,096 test points. Three models share two hidden layers, Fourier features, 30k Adam iterations and 400 sampling steps: the intrinsic construction at $c = 2 > n - 1$, the same at $c = 0.25 < n - 1$, and a tangent space baseline inspired by the hyperbolic wrapped normal [11]; in our implementation this is an OU diffusion in $\log_o$ coordinates with exact denoising targets. We report unbiased MMD² with an RBF kernel on hyperbolic distance, total variation of leaf weights after nearest leaf assignment, and the fraction of samples with $r > 6$, over three sampling seeds.

Two forward process diagnostics need no trained network: the prior mismatch, the unbiased MMD² between the forward endpoint cloud at time T and an independent sample of μ_c , and the fourth angular order parameter $R_4(t) = |\mathbb{E} e^{4i\varphi_t}|$, which measures the four fold group structure of the data and, by Proposition 3.3(ii), relaxes no more slowly than the gap.

Effect of the threshold. With architecture, features, budget and sampler fixed, changing c from 2.0 to 0.25 increases MMD² by a factor of 13, about seven times the combined sampling standard deviation, and roughly triples the leaf weight error (Table 2; samples in Figure 2, Appendix D). The forward process shows the problem before any network is trained. At $T = 4$ the $c = 2.0$ process is mixed at the resolution of the diagnostics: the prior mismatch is zero within estimator noise and R_4 has fallen from 0.765 to 0.010, the Monte Carlo floor. At $c = 0.25$ the radius has moved most of the

Table 2: Hierarchical generation on $\mathbb{H}^2$ (mean $\pm$ standard deviation over 3 sampling seeds; lower is better). Crossing the threshold with everything else fixed increases MMD^2 by a factor of 13; above it the intrinsic model is close to the wrapped baseline, which uses exact targets. The last two rows are forward process diagnostics at $T = 4$ on the 8,192 training trajectories and need no trained network.

	intrinsic, $c = 2.0 > n-1$	intrinsic, $c = 0.25 < n-1$	tangent (wrapped)
$\text{MMD}^2 (\times 10^3)$	0.50 ± 0.31	6.52 ± 0.77	0.64 ± 0.18
leaf TV	0.040 ± 0.010	0.128 ± 0.008	0.050 ± 0.007
diverged	0.0000	0.0000	0.0000
prior mismatch $\text{MMD}^2(p_T, \mu_c)$	≈ 0 (-1×10^{-5})	1.59×10^{-2}	n/a
forward $R_4(T)$ (data: 0.765)	0.010	0.380	n/a

way to equilibrium ($\mathbb{E}r = 3.91$ from 3.5, against an equilibrium mean of 4.19) while $R_4(T) = 0.380$ and the prior mismatch is 1.59×10^{-2} . Proposition 3.3 quantifies the separation: radial observables relax at $\lambda_1(c) + c = 0.29$ at $c = 0.25$, so $T = 4$ covers about one radial relaxation time, whereas the angular gap 0.038 corresponds to a time scale of 26.

Parameterisation ablation. Table 10 in Appendix D compares the score representations at $c = 2.0$ with single span targets and reduced budgets. In the budget matched 6k iteration comparison, anchoring with an extrinsic residual still sends 27.1% of samples beyond $r = 6$, whereas the frame representation sends none and reduces MMD^2 from 0.0452 to 0.0168; Fourier features, multi-span targets and 30k iterations then bring it to 0.0009.

Scope. The generative experiments use $n = 2$, one synthetic hierarchy and CPU sized networks. The wrapped baseline and the $c = 2$ intrinsic model have similar MMD^2 at the scale of the sampling variation, plausibly because the data occupy a bounded range of radii where the $\log_o$ chart distorts little, and the baseline has exact targets; we do not claim that the intrinsic model outperforms it. The experiments support four narrower conclusions: simulated rates reproduce the spectral gap of the forward generator; with architecture and protocol fixed, the confinement changes generative quality across the threshold; slow mixing below the threshold is visible in the forward process before training; and the frame representation removes the divergence seen with extrinsic outputs.

6 Limitations and open problems

Our lower bound analysis stops at the threshold: Theorem 3.5 bounds the small c gap from above and the exponent fits indicate that $\beta(n)$ is sharp, but a matching lower bound is open. Proposition 3.3 reduces it to one dimension, $\lambda_1(c) = \lambda^{(0)}(c) - c$ with $\lambda^{(0)}(c)$ the gap of the radial Sturm–Liouville problem with weight ρ_c , so the sharp exponent is the second order term in the small c expansion of that radial gap; the tightness of $\text{Var}_{\mu_c}(r)^{-1} - c$ (ratio 0.996 at $c = 0.25$ for $n = 4, 5$, Table 6) suggests that a variance expansion may reach it. Above the threshold, the expansion of Appendix B.7 gives the constant $-\frac{2}{3}(n-1)$ in $\lambda_1(c) - c$ and the $1/c$ coefficient $4(n-1)(n+2)/45$, both confirmed numerically to four digits, but it is formal and a proof along the lines of the harmonic approximation [22] is missing. A direct link between $\lambda_1(c)$ and generative error would need control of mixing, score estimation error and the bias of the multi-span target, perhaps through a manifold Girsanov bound. The generative experiments use $n = 2$ and synthetic data; real taxonomies such as WordNet [12] need the higher dimensional tangential head and larger models, and other geometries (the SPD cone, stratified tree spaces) need their own confinement or boundary mechanism.

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A Invariant measures across geometries: statements, proofs and tables

This appendix proves Theorem 2.1 and records the classification that places $\mathbb{H}^n$ among the geometries used in diffusion modelling, under the unit Laplacian convention $P_t = e^{t\Delta}$ for Brownian motion. The organising object is the symmetry group, following the viewpoint of Klein’s Erlangen programme [23, 24]. Once symmetry has identified the natural measure and spectral basis, three structural questions remain: is the volume finite, is there a boundary and is the state space continuous? Table 3 gives the short version; Tables 4 and 5 apply the classification to eleven spaces.

Proposition A.1 (Volume is stationary). *On a manifold without boundary, vol_g is infinitesimally invariant for Brownian motion. If Brownian motion is conservative, vol_g is invariant for the semigroup.*

Proof. For $f \in C_c^\infty(M)$, the divergence theorem gives $\int_M \Delta f \, d\text{vol}_g = \int_M \text{div}(\nabla f) \, d\text{vol}_g = 0$, which is infinitesimal invariance. If Brownian motion is conservative, symmetry of the heat semigroup gives $\int_M P_t f \, d\text{vol}_g = \int_M f P_t 1 \, d\text{vol}_g = \int_M f \, d\text{vol}_g$. $\square$

Proof of Theorem 2.1. (i) Proposition A.1, normalisation and the assumed conservativity give invariance of $\bar{\mu}$. Brownian motion on a connected manifold is irreducible: its heat kernel $p_t(x, y)$ is strictly positive for $t > 0$ and $x, y \in M$, by the strong maximum principle for the heat operator $\partial_t - \Delta$ [27]. The heat kernel is smooth, so the semigroup is strong Feller as well as irreducible; the standard strong Feller–irreducibility criterion therefore gives uniqueness of the invariant probability measure [28].

(ii) On a compact manifold $-\Delta$ has compact resolvent, hence discrete spectrum [26] with an $L^2(\text{vol}_g)$ -orthonormal eigenbasis $\{\phi_k\}_{k \geq 0}$, $-\Delta \phi_k = \lambda_k \phi_k$, $\lambda_0 = 0$ (the constants), $\lambda_1 > 0$. Writing $f = \sum_k c_k \phi_k$ with $c_0 = \int f \, d\bar{\mu}$, we have $P_t f = \sum_k e^{-\lambda_k t} c_k \phi_k$, and therefore $\|P_t f - c_0\|_{L^2}^2 = \sum_{k \geq 1} e^{-2\lambda_k t} |c_k|^2 \leq e^{-2\lambda_1 t} \sum_{k \geq 1} |c_k|^2$; take square roots.

(iii) A constant density has infinite total mass, which prevents vol_g from being normalised. For the drifted process, $\mathcal{L}_V$ is symmetric with respect to $d\mu_V = e^{-V} d\text{vol}_g$, since $e^V \text{div}(e^{-V} \nabla f) = \Delta f - \nabla V \cdot \nabla f = \mathcal{L}_V f$. Hence, for $f \in C_c^\infty(M)$, $\int_M \mathcal{L}_V f \, d\mu_V = \int_M \text{div}(e^{-V} \nabla f) \, d\text{vol}_g = 0$.

Together with the assumed conservativity, this gives invariance; irreducibility gives uniqueness as in (i). See [29] for the general theory. On $\mathbb{R}^n$, choosing $V(x) = \frac{c}{2}|x|^2$ gives $\mu_V = \mathcal{N}(0, c^{-1}I)$, the stationary law of $dX = -cX dt + \sqrt{2} dB$. $\square$

Proposition A.2 (Drift depends on volume; the rate is a separate, spectral matter). *Let (M, g) be connected, complete and without boundary.*

- (a) *If $\text{vol}_g(M) < \infty$, whether or not M is compact, the normalised volume is the unique invariant probability measure of Brownian motion, with no drift.*
- (b) *Convergence is exponential at rate λ_1 , the bottom of the nonzero spectrum of $-\Delta$. Compactness guarantees $\lambda_1 > 0$. Finite volume alone does not give the compact spectral picture. For example, a hyperbolic surface of finite area that is not compact has essential spectrum $[1/4, \infty)$. Across compact hyperbolic surfaces, the positive gap can be arbitrarily small [26].*

Proof. Part (a) is Theorem 2.1(i). Its proof uses only finiteness of volume and irreducibility. Part (b) is the estimate of Theorem 2.1(ii), which requires only $\lambda_1 > 0$. If $\Gamma \backslash \mathbb{H}^2$ is not compact but has finite area, the essential spectrum of $-\Delta$ is $[1/4, \infty)$ and there are at most finitely many eigenvalues below it [26]. The normalised volume is therefore stationary even though the spectrum is not discrete. $\square$

Corollary A.3 (Symmetry–volume spine). *Suppose $M = G/H$ carries a G -invariant Riemannian metric.*

- (a) *vol_g is G -invariant and is the unique G -invariant measure up to positive scale; it is invariant for Brownian motion.*
- (b) *If G is compact then M is compact; the normalised G -invariant measure is the unique stationary law and no drift is needed. When $M = G$ is a compact Lie group with a bi-invariant metric, this measure is Haar measure.*
- (c) *If M is non-compact with $\text{vol}_g(M) = \infty$, the G -invariant measure is not normalisable and a prior therefore requires a drift $-\nabla V$; the resulting μ_V is invariant only under $\{\gamma \in G : V \circ \gamma = V\}$, which necessarily breaks full G -invariance.*

Proof. (a) G acts by isometries and therefore preserves vol_g ; transitivity makes any two G -invariant Radon measures proportional. (b) $M = G/H$ is the continuous image of the compact group G , hence compact; apply Theorem 2.1(i)–(ii) and identify the bi-invariant volume on a compact group with Haar measure. (c) The G -invariant measure is vol_g up to scale, and it is infinite by hypothesis. Theorem 2.1(iii) therefore forces a drift. An isometry γ preserves $\mu_V \propto e^{-V} \text{vol}_g$ if and only if $V \circ \gamma = V$. $\square$

On $\mathbb{H}^n$, the sheet preserving isometry group $O^+(n, 1)$ preserves the infinite hyperbolic volume, whereas the drift $-c\nabla(\frac{1}{2}r^2)$ selects a base point and reduces the symmetry to the stabiliser $O(n)$. Any normalisable radial prior of this form therefore introduces a preferred centre. It is this residual $O(n)$ symmetry that produces the sector decomposition used in Section 3.3.

Proposition A.4 (Boundary flux and Neumann reflection). *Let M have boundary ∂M with outward unit normal ν . For $f \in C^\infty(\bar{M})$, $\int_M \Delta f d\text{vol}_g = \int_{\partial M} \partial_\nu f dS$. Consequently vol_g is invariant only if the boundary flux vanishes. Reflected (Neumann) Brownian motion has vol_g as an invariant measure, its generator acting on functions with $\partial_\nu f = 0$ on ∂M . Equivalently, a diffusion whose coefficients degenerate at ∂M keeps vol_g stationary without reflection, since the process never reaches the boundary.*

Proof. The identity is Green’s theorem on a manifold with boundary. Stationarity for vol_g requires $\int_M \Delta f d\text{vol}_g = 0$ for all admissible f . By the identity this holds precisely when the boundary integral vanishes, which the Neumann condition enforces. The degenerate alternative is exemplified

by the Wright–Fisher (Jacobi) diffusion on the simplex, whose diffusion coefficient vanishes on each face and keeps trajectories in the interior. $\square$

Remark A.5 (No manifold, no SDE). When the state space is a finite or countable set S , the generator Δ is replaced by a Markov rate matrix Q and stationarity becomes $\pi Q = 0$. A group Γ acting on S is a symmetry when Q commutes with its permutation representation. Curvature, volume and boundary have no direct analogue; the relevant distinction is smooth versus discrete, with equivariance imposed on the model rather than realised by a metric.

Proposition A.6 (Spectral expansion and Peter–Weyl). *On a compact manifold, $-\Delta$ has discrete spectrum with an orthonormal eigenbasis $\{\phi_k\}$, and the heat kernel is $p_t(x, y) = \sum_k e^{-\lambda_k t} \phi_k(x) \overline{\phi_k(y)}$. Suppose further that $M = G/H$ is a compact homogeneous space whose metric is invariant under G . Then Δ commutes with the G action. Peter–Weyl theory [30] decomposes $L^2(G/H)$ into irreducible G representations that contain vectors fixed by H , and Δ acts blockwise on this decomposition. Diagonalising within the finite multiplicity blocks gives a representation theoretic eigenbasis. In the classical multiplicity free cases this recovers spherical harmonics on S^n , Wigner D functions on $\text{SO}(3)$ and Fourier characters on the torus.*

Proof. The spectral expansion is standard on compact manifolds. Since G acts by isometries, Δ commutes with the induced unitary representation and therefore preserves each isotypic component of $L^2(G/H)$. Peter–Weyl identifies the components that occur with the irreducible representations admitting vectors fixed by H . The self adjoint operator $-\Delta$ can then be diagonalised on the corresponding finite multiplicity spaces, yielding the stated eigenbasis and the familiar bases in the listed examples. $\square$

Remark A.7 (Curvature feeds compactness: Bonnet–Myers). If (M, g) is complete and $\text{Ric} \geq (n-1)\kappa g$ with $\kappa > 0$, then M is compact with diameter at most $\pi/\sqrt{\kappa}$ [25]. A strictly positive Ricci lower bound therefore forces the finite volume regime with no drift, whereas the non-compact examples $\mathbb{R}^n$ and $\mathbb{H}^n$ cannot satisfy such a bound. On $\mathbb{H}^n$, curvature sets the confinement strength needed by the Bakry–Émery estimate, which is the role studied in the main text.

Table 3: The classification in compact form. Symmetry supplies the prior and the basis; finite volume decides the drift; a boundary decides reflection; discreteness leaves the SDE regime. The $\mathbb{H}^n$ row is this paper.

Space	vol $< \infty$	Stationary prior	Forward process	Kernel basis
$\mathbb{R}^n$	no	Gaussian (via drift)	OU/VP SDE	Hermite; Fourier
$S^n, \text{SO}(3), T^n$	yes	uniform/Haar	Brownian motion	harmonics; Wigner D ; Fourier
$\text{SE}(3)$	no	Haar $\times$ Gaussian	BM $\times$ OU, factorised	Wigner $D \times$ Gaussian
simplex Δ^{n-1}	yes	uniform (Dirichlet)	Wright–Fisher, degenerate at the boundary	Jacobi polynomials
$\mathbb{H}^n$	no	$\mu_c \propto e^{-cr^2/2} d\text{vol}$ (§3)	drifted BM $\mathcal{L}_c$ (§3)	radial $\times$ spherical harmonics
SPD cone	no	drift dependent	affine invariant BM + drift	zonal functions
discrete	n/a	uniform/absorbing	continuous time Markov chain	matrix exponential
BHV tree space	no	open	open	n/a

Tables 4 and 5 apply the same classification to eleven spaces in a common row order. Table 4 records the geometry. Table 5 records the corresponding diffusion choices. The stationary measure entries follow Corollary A.3. The obstruction column uses Theorem 2.1, Proposition A.4 and Remark A.5. The spectral column follows Proposition A.6. The $\mathbb{H}^n$ row is established in the main text.

Table 4: Geometric profile. What each space is: curvature, compactness, boundary, topology and the symmetry group that organises the rest. Row order matches Table 5.

Space	Curvature	Compact?	Boundary	Topology	Symmetry group
Euclidean space $\mathbb{R}^n$	Flat ($K = 0$)	No	None	Contractible; π_1 trivial	$E(n) = O(n) \ltimes \mathbb{R}^n$ (flat homogeneous)
n -sphere $S^n \subset \mathbb{R}^{n+1}$	Positive ($K = +1$)	Yes	None	Simply connected ($n \geq 2$); $\pi_1(S^1) = \mathbb{Z}$	$O(n+1)$; $S^n \cong O(n+1)/O(n)$
Flat torus $T^n = \mathbb{R}^n/\mathbb{Z}^n$	Flat ($K = 0$)	Yes	None	$\pi_1 = \mathbb{Z}^n$ (not simply connected)	Abelian T^n (translations)
Rotations $SO(3)$ (compact Lie group)	Non-negative (bi-invariant)	Yes	None	$\pi_1 = \mathbb{Z}_2$; $SO(3) \cong \mathbb{RP}^3$ (cover $SU(2)$)	Left/right translation; bi-invariant metric
Rigid motions $SE(3) = SO(3) \ltimes \mathbb{R}^3$	Mixed; no bi-invariant metric	No	None	$\pi_1 = \mathbb{Z}_2$; $\mathbb{R}^3$ contractible	Left-invariant; semidirect product
Hyperbolic space $\mathbb{H}^n$	Negative ($K = -1$)	No	None (∂_∞ not a manifold boundary)	Contractible (Hadamard $\cong \mathbb{R}^n$)	$O^+(n, 1)$; $\mathbb{H}^n \cong SO^+(n, 1)/SO(n)$
SPD cone $\text{Sym}^+(n)$ (affine-invariant metric)	Non-positive (Hadamard/CAT(0))	No	None intrinsically (singular PSD at ∞)	Contractible (open convex cone)	$GL(n)$ congruence $P \mapsto APA^\top$; $\cong GL(n)/O(n)$
Probability simplex Δ^{n-1}	Positive (Fisher–Rao)	Yes	Yes: faces where some $x_i = 0$	Contractible (convex); faces form boundary	S_n permutations; $\sqrt{\cdot}$ -isometry to S^{n-1} orthant
Stiefel $V_k(\mathbb{R}^n)$; Grassmann $Gr(k, n)$	Non-negative (compact symmetric)	Yes	None	Non-trivial (e.g. $\pi_1(Gr) = \mathbb{Z}_2$, real)	$O(n)$ acts transitively; homogeneous
Graphs/categorical discrete data	N/A (not a smooth manifold)	N/A (discrete)	N/A	Combinatorial; no differential structure	S_n (node permutations/automorphisms)
Stratified/BHV tree space $\mathcal{T}_n$	CAT(0), singular	No	Strata (collapsed edges)	Cone-complex; not a manifold	Label permutations; local orthant charts

Table 5: Diffusion prescription. What geometry forces on the process: admissible forward dynamics, the stationary prior, the obstruction and its remedy, the natural kernel basis and representative methods. Row order matches Table 4.

Space	Forward process	Stationary prior	Obstruction & remedy	Kernel/eigenbasis	Representative methods
Euclidean space $\mathbb{R}^n$	VP/OU SDE (mean-reverting); or VE (variance-exploding)	Isotropic Gaussian $\mathcal{N}(0, I)$	Non-compact $\rightarrow$ drift needed for a normalisable prior	Gaussian heat kernel; Hermite/Fourier	DDPM; Score-SDE
n -sphere S^n	Intrinsic heat flow/Riemannian BM (Laplace–Beltrami)	Uniform (surface measure)	None (compact): no drift required	Spherical harmonics $Y_{\ell m}$	Riemannian SGM; directional data
Flat torus T^n	Wrapped BM (heat kernel = wrapped Gaussian)	Uniform on T^n	None (compact); must respect periodicity	Fourier modes $e^{ik \cdot x}$, $k \in \mathbb{Z}^n$	Torsional diffusion (conformers)
Rotations $\text{SO}(3)$	Group BM = IGSO(3) (its heat kernel)	Haar measure (= uniform)	None (compact group): no drift	Wigner- D functions D_{mn}^ℓ	FrameDiff; RFdiffusion
Rigid motions $\text{SE}(3)$	Factorised: BM on $\text{SO}(3) \times \text{OU}$ on $\mathbb{R}^3$	$\text{Haar}(\text{SO}(3)) \times \mathcal{N}(\mathbb{R}^3)$	Non-compact translation $\rightarrow$ drift; rotation needs none	Wigner- $D \times$ Gaussian	SE(3) diffusion (poses, docking, robotics)
Hyperbolic space $\mathbb{H}^n$	Confined BM $\mathcal{L}_c$; other Riemannian drifts possible	$\mu_c \propto e^{-cr^2}/2 \, d\text{vol}$ for this confinement	Non-compact, $K < 0 \rightarrow$ no Brownian stationary probability; $c > n - 1$ gives the Bakry–Émery rate	Radial functions $\times$ spherical harmonics; heat kernel (McKean)	This paper; Riemannian diffusion; wrapped normal
SPD cone $\text{Sym}^+(n)$	Affine-invariant BM; log-Euclidean/Wishart-type	Log-normal/Wishart-type (via drift)	Non-compact $\rightarrow$ drift; iterates must stay PD	Zonal/matrix spherical functions	Covariance; DTI tensors; connectivity
Probability simplex Δ^{n-1}	Wright–Fisher/Jacobi; or $\sqrt{\cdot}$ -map to sphere; or reflected BM	$\text{Dirichlet}(1, \dots, 1) = \text{uniform}$	Boundary $\rightarrow$ reflection or a face-preserving process	Jacobi polynomials; harmonics via $\sqrt{\cdot}$ -map	Compositional data; Dirichlet diffusion
Stiefel/Grassmann	Riemannian BM on the homogeneous space	Uniform ($O(n)$ -invariant measure)	None (compact); retract to enforce constraint	Zonal/representation-theoretic functions	Frames, subspaces, PCA-type constraints
Graphs/discrete	CTMC noising (uniform/absorbing/marginal): not an SDE	Uniform over categories, or δ at absorbing [MASK]	Discrete $\rightarrow$ Markov jump; impose permutation equivariance	Transition-matrix exponential	D3PM; DiGress
Stratified/tree space	No canonical BM across strata (candidates: sticky/reflected)	Open: no agreed invariant law	Singular strata $\rightarrow$ manifold diffusion undefined	n/a	Frontier/open problem

B Proofs for Section 3

B.1 Proof of Theorem 3.1

Proof of Theorem 3.1. (i) In geodesic polar coordinates about o , $d\text{vol} = \sinh^{n-1}(r) dr d\sigma$. Since $\sinh^{n-1}(r) \leq Ce^{(n-1)r}$ whereas $e^{-cr^2/2}$ has Gaussian decay, μ_c has finite mass. The generator is symmetric in $L^2(\mu_c)$ and can be written as $\mathcal{L}_c = e^{W_c} \text{div}(e^{-W_c} \nabla \cdot)$. It follows that $\int \mathcal{L}_c f d\mu_c = \int \text{div}(e^{-W_c} \nabla f) d\text{vol} = 0$, which proves invariance of μ_c . The heat kernel of this uniformly elliptic diffusion is smooth and strictly positive on the connected manifold, so the semigroup is strong Feller and irreducible; uniqueness follows from the same criterion [28]. Finally, r^2 is a Lyapunov function for the confining drift and rules out explosion. (ii) On $\mathbb{H}^n$, $\text{Ric} = -(n-1)g$, while the Hessian comparison theorem on a Cartan–Hadamard manifold gives $\nabla^2(\frac{1}{2}r^2) \geq g$ [25]. Hence $\nabla^2 W_c \geq cg$ and $\text{Ric} + \nabla^2 W_c \geq (c - (n-1))g$, after which the Bakry–Émery criterion yields the Poincaré inequality with constant $c - (n-1)$ and the stated decay [29]. $\square$

Throughout, $d\mu_c = Z_c^{-1} \sinh^{n-1}(s) e^{-cs^2/2} ds d\sigma(\theta)$ in geodesic polar coordinates, with σ the normalised measure on S^{n-1} and $Z_c = \int_0^\infty \sinh^{n-1}(s) e^{-cs^2/2} ds$. We use repeatedly that for $s \geq 1$,

$$\gamma e^s \leq \sinh s \leq \frac{1}{2}e^s, \quad \gamma := \frac{1}{2}(1 - e^{-2}), \quad (4)$$

and that $\sinh x \leq x e^{x^2/6}$ for all $x \geq 0$ (compare the series termwise: $(2k+1)! \geq 6^k k!$).

B.2 Proof of Theorem 3.2

The Rayleigh bound. The function $f = r$ is 1-Lipschitz with $|\nabla r| = 1$ away from o , lies in $L^2(\mu_c)$ and belongs to the form domain of $\mathcal{L}_c$ (Lipschitz functions do; see [29]). The variational characterisation gives

$$\lambda_1(c) \leq \frac{\int |\nabla r|^2 d\mu_c}{\text{Var}_{\mu_c}(r)} = \frac{1}{\text{Var}_{\mu_c}(r)}.$$

The radial variance at strong confinement. Under μ_c the radius has density $\rho_c(s) = Z_c^{-1} \sinh^{n-1}(s) e^{-cs^2/2}$ on $(0, \infty)$. Let $U = \sqrt{c}r$, with density proportional to

$$h_c(u) = c^{(n-1)/2} \sinh^{n-1}(u/\sqrt{c}) e^{-u^2/2}.$$

Pointwise $h_c(u) \rightarrow u^{n-1} e^{-u^2/2}$ as $c \rightarrow \infty$, and by $\sinh x \leq x e^{x^2/6}$,

$$h_c(u) \leq u^{n-1} e^{(n-1)u^2/(6c)} e^{-u^2/2} \leq u^{n-1} e^{-u^2/3} \quad (c \geq n-1),$$

an integrable dominating function that remains integrable against u and u^2 . Dominated convergence applied to the normalisation and to the first two moments gives $\mathbb{E}U \rightarrow \mathbb{E}\chi_n$ and $\mathbb{E}U^2 \rightarrow \mathbb{E}\chi_n^2 = n$, where χ_n has density proportional to $u^{n-1} e^{-u^2/2}$ and $\mathbb{E}\chi_n = \sqrt{2} \Gamma(\frac{n+1}{2}) / \Gamma(\frac{n}{2})$. Hence

$$c \text{Var}_{\mu_c}(r) = \text{Var}(U) \longrightarrow v_n = n - 2 \left(\frac{\Gamma((n+1)/2)}{\Gamma(n/2)} \right)^2.$$

Positivity of v_n holds because χ_n is not deterministic. For the upper bound $v_n \leq 1$, write $\chi_n = |Z|$ with $Z \sim \mathcal{N}(0, I_n)$. The map $z \mapsto |z|$ is 1-Lipschitz, and the Gaussian Poincaré inequality gives $\text{Var}(|Z|) \leq \mathbb{E}|\nabla |z||^2 = 1$.

Conclusion. Choose $c_0(n)$ such that $\text{Var}_{\mu_c}(r) \geq v_n/(2c)$ for $c \geq c_0(n)$. Then $\lambda_1(c) \leq 2c/v_n$, which with Theorem 3.1(ii) sandwiches $\lambda_1(c)$ between two linear functions of c . $\square$

B.3 Proof of Proposition 3.3

Sector decomposition. The rotations of $\mathbb{H}^n$ about o act by isometries that fix r , so they commute with $\mathcal{L}_c$ and preserve μ_c . Expanding $f(s, \theta) = \sum_\ell \sum_j g_{\ell j}(s) Y_{\ell j}(\theta)$ in an orthonormal basis of

spherical harmonics, with $-\Delta_{S^{n-1}} Y_{\ell j} = \ell(\ell + n - 2) Y_{\ell j}$, the Dirichlet form separates:

$$\int |\nabla f|^2 d\mu_c = \sum_{\ell, j} \int_0^\infty \left(g_{\ell j}^2 + \frac{\ell(\ell + n - 2)}{\sinh^2 s} g_{\ell j}^2 \right) \rho_c ds = \sum_{\ell, j} \mathcal{E}_\ell(g_{\ell j}),$$

since $|\nabla f|^2 = (\partial_s f)^2 + \sinh^{-2}(s) |\nabla_{S^{n-1}} f|^2$ in polar coordinates. Hence $-\mathcal{L}_c$ is the orthogonal sum of the operators $-\mathcal{L}_c^{(\ell)}$ of (1), each acting on $\mathcal{H} = L^2((0, \infty), \rho_c ds)$ and repeated with the multiplicity $\dim \mathcal{Y}_\ell$ of the harmonics of degree ℓ . Part (ii) follows immediately, because $\mathcal{E}_\ell \geq \mathcal{E}_1$ on a common form domain for every $\ell \geq 1$.

Factorisation. Let $D = \partial_s$ on $\mathcal{H}$ with domain $\{F \in \mathcal{H} : F \text{ locally absolutely continuous, } F' \in \mathcal{H}\}$, which is the radial part of the form domain of $\mathcal{L}_c$. Then D is closed and densely defined, $\|DF\|^2 = \mathcal{E}_0(F)$, and the self adjoint operator associated with this closed form is $D^*D = -\mathcal{L}_c^{(0)}$. For $g \in C_c^\infty((0, \infty))$, integrating by parts against the weight ρ_c and using $\rho'_c = a\rho_c$ gives

$$\langle DF, g \rangle_{\mathcal{H}} = \int_0^\infty F' g \rho_c ds = - \int_0^\infty F (g' + ag) \rho_c ds, \quad \text{so} \quad D^*g = -g' - ag.$$

For $g \in \text{dom}(D^*)$, a second integration by parts on $[\varepsilon, \infty)$ gives

$$\|D^*g\|^2 = \lim_{\varepsilon \rightarrow 0} \int_\varepsilon^\infty (g'^2 + a^2 g^2 + a(g^2)') \rho_c ds = \lim_{\varepsilon \rightarrow 0} \left(\int_\varepsilon^\infty (g'^2 - a' g^2) \rho_c ds + [a g^2 \rho_c]_\varepsilon^\infty \right),$$

because $(a\rho_c)' = (a' + a^2)\rho_c$. Since $-a'(s) = (n-1)\sinh^{-2}s + c$, the integrand is $g'^2 + (n-1)\sinh^{-2}(s)g^2 + cg^2$. The boundary term at infinity vanishes along a sequence $s_k \uparrow \infty$, because $|a(s)| \leq Cs$ for large s and $\int_0^\infty g^2 \rho_c ds < \infty$. The boundary term at the origin satisfies $a\rho_c = \rho'_c \sim (n-1)Z_c^{-1}s^{n-2}$ as $s \rightarrow 0$, and it vanishes along a sequence $\varepsilon_k \downarrow 0$ for every g with $\int_0 g^2 \sinh^{-2}(s) \rho_c ds < \infty$, which is the finiteness of $\mathcal{E}_1(g)$. Letting $\varepsilon \rightarrow 0$ along this sequence,

$$\|D^*g\|^2 = \mathcal{E}_1(g) + c\|g\|^2, \quad \text{that is} \quad DD^* = -\mathcal{L}_c^{(1)} + c,$$

where the identification of $\text{dom}(D^*)$ with the form domain of $\mathcal{E}_1$ uses that $C_c^\infty((0, \infty))$ is a core for both forms; the origin is a polar set for $n \geq 2$.

Isospectrality. For a closed densely defined operator A between Hilbert spaces, the polar decomposition $A = U|A|$ makes A^*A restricted to $(\ker A)^\perp$ unitarily equivalent to AA^* restricted to $(\ker A^*)^\perp$; in particular the nonzero spectra of A^*A and AA^* coincide with multiplicity [21]. Here $\ker D$ consists of the constants, which carry the eigenvalue 0 of $-\mathcal{L}_c^{(0)}$, and $\ker D^* = \{g : (g\rho_c)' = 0\} = \{C/\rho_c\} \cap \mathcal{H} = \{0\}$ because $\int_0^\infty \rho_c^{-1} \rho_c ds = \infty$. Therefore the nonzero spectrum of $-\mathcal{L}_c^{(0)}$ equals the spectrum of $-\mathcal{L}_c^{(1)} + c$, which is part (i). The eigenfunction statement is the same identity at the level of solutions: differentiating $F'' + aF' = -\Lambda F$ gives $(F')'' + a(F')' + a'F' = -\Lambda F'$, that is $-\mathcal{L}_c^{(1)}F' = (\Lambda - c)F'$ by the expression for $-a'$. Part (iii) combines (i), (ii) and $c > 0$: the bottom of the spectrum of $-\mathcal{L}_c$ on the orthogonal complement of the constants is $\min\{\lambda^{(1)} + c, \lambda^{(1)}, \lambda^{(2)}, \dots\} = \lambda^{(1)}$. $\square$

On $\mathbb{R}^n$ the same computation with $\rho_c(s) \propto s^{n-1}e^{-cs^2/2}$ gives $-a' = (n-1)s^{-2} + c$ and reproduces the OU ladder, in which the degree one sector has bottom c and the radial sector has bottom $2c$. On $\mathbb{H}^n$ the map $F \mapsto F'\theta_1$ survives, but the degree one bottom is no longer c ; Theorem 3.5 shows that it is exponentially small below the threshold. Since $-(\log \rho_c)'' = -a' \geq c$, the radial density is log-concave with constant c , so the one dimensional Bakry–Émery argument gives $\lambda^{(0)}(c) \geq c$ for every $c > 0$, in agreement with $\lambda^{(0)}(c) = c + \lambda_1(c)$ and $\lambda_1(c) > 0$.

Proof of Corollary 3.4. The test function r lies in the radial sector, so the Rayleigh bound of Theorem 3.2 applies to $\lambda^{(0)}(c)$; subtracting c by Proposition 3.3(iii) gives the first inequality. The $\liminf$ follows from Theorem 3.1(ii) and the $\limsup$ from $c\text{Var}_{\mu_c}(r) \rightarrow v_n$. The decay statements follow from the sector decomposition and from (iii). $\square$

543 **B.4 Proof of Theorem 3.5**

544 **Test function.** Let $\psi(\theta) = \sqrt{n} \theta_1$ on S^{n-1} (for $n = 2$, $\psi(\varphi) = \sqrt{2} \cos \varphi$): a first spherical
545 harmonic with

$$\int \psi d\sigma = 0, \quad \int \psi^2 d\sigma = 1, \quad \int |\nabla_{S^{n-1}} \psi|^2 d\sigma = n - 1,$$

546 the last since ψ is an eigenfunction of the spherical Laplacian with eigenvalue $n - 1$. Let η be a fixed
547 smooth cutoff with $\eta = 0$ on $[0, 1]$, $\eta = 1$ on $[2, \infty)$, $0 \leq \eta \leq 1$, $|\eta'| \leq 2$, and set $f(x) = \eta(r)\psi(\theta)$:
548 smooth and bounded with bounded gradient, since $|\nabla f|^2 = \eta'(r)^2 \psi^2 + \eta(r)^2 |\nabla_S \psi|^2 / \sinh^2 r$ and
549 η vanishes near the origin.

550 **Mean and variance.** $\int f d\mu_c = (Z_c^{-1} \int \eta \sinh^{n-1} e^{-cs^2/2} ds) (\int \psi d\sigma) = 0$, and

$$\text{Var}_{\mu_c}(f) = Z_c^{-1} \int_1^\infty \eta^2 \sinh^{n-1}(s) e^{-cs^2/2} ds \geq \mu_c^{\text{rad}}([2, \infty)).$$

551 We claim $\kappa_n := \inf_{0 < c \leq 1} \mu_c^{\text{rad}}([2, \infty)) > 0$. Indeed $c \mapsto \mu_c^{\text{rad}}([2, \infty))$ is continuous and strictly
552 positive on $(0, 1]$ (dominated convergence in both numerator and Z_c), while as $c \downarrow 0$,

$$\mu_c^{\text{rad}}([0, 2]) \leq Z_c^{-1} \int_0^2 \sinh^{n-1}(s) ds \leq Z_c^{-1} e^{2(n-1)} \rightarrow 0$$

553 by the lower bound (5) on Z_c below, and hence the function tends to 1 at the open end. A positive
554 continuous function on $(0, 1]$ with positive limit inferior at 0 has positive infimum.

555 **Energy.** Using the two identities for ψ ,

$$\begin{aligned} \int |\nabla f|^2 d\mu_c &= Z_c^{-1} \left[\int_1^2 \eta'^2 \sinh^{n-1} e^{-cs^2/2} ds + (n-1) \int_1^\infty \eta^2 \sinh^{n-3}(s) e^{-cs^2/2} ds \right] \\ &\leq Z_c^{-1} [4e^{2(n-1)} + (n-1)A_c], \end{aligned}$$

556 where $A_c := \int_1^\infty \sinh^{n-3}(s) e^{-cs^2/2} ds$ and we bounded $\int_1^2 \sinh^{n-1} \leq \sinh^{n-1}(2) \leq e^{2(n-1)}$.

557 **The partition function from below.** By (4), for $0 < c \leq 1$,

$$Z_c \geq \gamma^{n-1} \int_1^\infty e^{(n-1)s - cs^2/2} ds = \gamma^{n-1} e^{\frac{(n-1)^2}{2c}} \int_1^\infty e^{-\frac{c}{2}(s - \frac{n-1}{c})^2} ds \geq \frac{1}{2} \gamma^{n-1} \sqrt{\frac{2\pi}{c}} e^{\frac{(n-1)^2}{2c}}, \quad (5)$$

558 since the Gaussian centre $(n-1)/c \geq n-1 \geq 1$ lies in the integration range, the integral contains
559 at least half the Gaussian mass.

560 **The angular integral from above.** For $n = 2$, (4) gives $\sinh^{-1}(s) \leq \gamma^{-1} e^{-s}$ on $[1, \infty)$, hence
561 $A_c \leq \gamma^{-1} e^{-1}$, a constant. For $n \geq 3$, $\sinh^{n-3}(s) \leq 2^{-(n-3)} e^{(n-3)s}$ and completing the square,

$$A_c \leq 2^{-(n-3)} \sqrt{\frac{2\pi}{c}} e^{\frac{(n-3)^2}{2c}}.$$

562 **Conclusion.** Combining, $\lambda_1(c) \leq \kappa_n^{-1} Z_c^{-1} [4e^{2(n-1)} + (n-1)A_c]$. For $n = 2$, both terms in the
563 bracket are constants and (5) gives

$$\lambda_1(c) \leq C \sqrt{c} e^{-\frac{1}{2c}}.$$

564 For $n \geq 3$, the cutoff term contributes $C_1(n) \sqrt{c} e^{-(n-1)^2/(2c)} \leq C_1(n) e^{-(n-1)^2/(2c)}$, while the
565 angular term contributes, after the $\sqrt{2\pi/c}$ factors cancel,

$$C_2(n) e^{-\frac{(n-1)^2 - (n-3)^2}{2c}} = C_2(n) e^{-\frac{2(n-2)}{c}},$$

566 whence $\lambda_1(c) \leq C(n) e^{-\beta(n)/c}$ with $\beta(n) = \min\{(n-1)^2/2, 2(n-2)\}$. $\square$

The proof makes the two costs of Section 3.4 literal. The first exponent is the relative mass of a unit ball at the origin against a shell at radius $(n-1)/c$. The second is the angular kinetic energy of the first harmonic on that shell, where the metric weight is $\sinh^{-2} s \approx 4e^{-2s}$. For $n=3$ the two exponents coincide at $2/c$. Since $\frac{(n-1)^2}{2} - 2(n-2) = \frac{(n-3)^2}{2} \geq 0$, the angular cost $2(n-2)$ is the smaller for every $n \geq 3$ and determines $\beta(n)$, with strict inequality once $n \geq 4$.

B.5 Proof of Lemma 4.1

Proof of Lemma 4.1. Using $\langle x, x \rangle_L = -1$, $|P_x w|_g^2 = \langle w, w \rangle_L + \langle x, w \rangle_L^2$. For $w = e_{n+1}$ this equals $-1 + \cosh^2 r = \sinh^2 r$, while in general $\langle w, w \rangle_L \leq |w|_2^2 = 1$. Cauchy–Schwarz, applied to w with its last coordinate negated, gives $\langle x, w \rangle_L^2 \leq |x|_2^2 |w|_2^2 = \cosh 2r$; hence $|P_x w|_g^2 \leq 1 + \cosh 2r = 2 \cosh^2 r$. $\square$

B.6 Positivity for every $c > 0$ (Remark 3.6)

Fix $\alpha > 0$ and let $V = \chi(r)e^{\alpha r}$ with χ a smoothing that makes V of class C^2 near o and $V \geq 1$. Away from o , using $\Delta r = (n-1) \coth r$ and $|\nabla r| = 1$,

$$\frac{\mathcal{L}_c V}{V} = \alpha^2 + \alpha(n-1) \coth r - \alpha c r \xrightarrow{r \rightarrow \infty} -\infty,$$

Thus there are $\theta > 0$, $b < \infty$ and $R < \infty$ with $\mathcal{L}_c V \leq -\theta V + b \mathbf{1}_{B_R}$. This Lyapunov condition, together with the local Poincaré inequality on balls, implies a global Poincaré inequality for μ_c by the criterion of [19]; hence $\lambda_1(c) > 0$ for every $c > 0$.

B.7 The large c expansion

Remark B.1 (Above the threshold: the Euclidean limit). For large c the law μ_c concentrates at scale $c^{-1/2}$ about o , where $\coth s = s^{-1} + s/3 + O(s^3)$ and $\sinh^{-2} s = s^{-2} - \frac{1}{3} + O(s^2)$. Inserting these expansions in (1) with $\ell = 1$ gives

$$-\mathcal{L}_c^{(1)} = \left[-\partial_s^2 - \left(\frac{n-1}{s} - \left(c - \frac{n-1}{3} \right) s \right) \partial_s + \frac{n-1}{s^2} \right] - \frac{n-1}{3} + \mathcal{E}_{\text{rem}},$$

where the bracket is the degree one sector of the Euclidean OU operator with confinement $c - (n-1)/3$, whose bottom eigenvalue is $c - (n-1)/3$ with eigenfunction s , and where $\mathcal{E}_{\text{rem}}$ collects the higher order terms, which are of relative size $O(1/c)$ at scale $c^{-1/2}$. Carrying the expansion to second order (derivation below) gives

$$\lambda_1(c) = c - \frac{2}{3}(n-1) + \frac{4(n-1)(n+2)}{45c} + O(c^{-2}), \quad \lambda^{(0)}(c) = \lambda_1(c) + c,$$

the second by Proposition 3.3(iii). The computation is short because s is an exact eigenfunction of the first order perturbation, so the second order coefficient is a Gaussian moment ratio. The expansion is formal: a rigorous version would follow the harmonic approximation for low lying eigenvalues [22], which we do not carry out, so we state it as a prediction. Numerically it holds. On $\mathbb{H}^n$ for $n = 2, \dots, 5$, the finite element values of $\lambda_1(c) - c$ at $c = 256$ are $-0.6653, -1.3299, -1.9938$ and -2.6569 against the predicted $-\frac{2}{3}(n-1) = -0.6667, -1.3333, -2$ and -2.6667 , and the scaled remainder $c(\lambda_1(c) - c + \frac{2}{3}(n-1))$ converges as $c \rightarrow \infty$ to $0.356, 0.889, 1.600$ and 2.489 , against $4(n-1)(n+2)/45 = 0.356, 0.889, 1.600$ and 2.489 . In particular $\lambda_1(c)/c \rightarrow 1$, the Bakry–Émery bound $c - (n-1)$ of Theorem 3.1(ii) is sharp to leading order with a deficit tending to $(n-1)/3$, and the upper end $v_n^{-1} - 1$ of the ratio interval in Corollary 3.4 is not attained.

Derivation. Substitute $s = y/\sqrt{c}$ in (1) with $\ell = 1$ and expand $\coth(y/\sqrt{c}) = \sqrt{c}/y + y/(3\sqrt{c}) - y^3/(45c^{3/2}) + O(c^{-5/2})$ and $\sinh^{-2}(y/\sqrt{c}) = c/y^2 - \frac{1}{3} + y^2/(15c) + O(c^{-2})$, with $\partial_s = \sqrt{c}\partial_y$. Collecting powers of c ,

$$-\mathcal{L}_c^{(1)} = cT_0 + T_1 + c^{-1}T_2 + O(c^{-2}),$$

604

$$T_0 = -\partial_y^2 - \left(\frac{n-1}{y} - y\right)\partial_y + \frac{n-1}{y^2}, \quad T_1 = -\frac{n-1}{3}(y\partial_y + 1), \quad T_2 = (n-1)\left(\frac{y^3}{45}\partial_y + \frac{y^2}{15}\right).$$

605 T_0 is the degree one sector of the Euclidean OU operator with unit confinement. It is symmetric
 606 on $L^2(w_0 dy)$ with $w_0(y) = y^{n-1}e^{-y^2/2}$, its lowest eigenvalue 1 is simple, and the eigenfunction
 607 is $g_0(y) = y$. Write $\lambda_1(c) = c\lambda^{[0]} + \lambda^{[1]} + c^{-1}\lambda^{[2]} + O(c^{-2})$ and expand the eigenfunction
 608 accordingly. Since T_1 and T_2 are not symmetric on $L^2(w_0 dy)$, the Rayleigh–Schrödinger coefficients
 609 are computed with the left eigenvector of T_0 in $L^2(dy)$: symmetry of T_0 on $L^2(w_0 dy)$ means that its
 610 $L^2(dy)$ adjoint is $w_0 T_0 w_0^{-1}$, so the left eigenvector for the eigenvalue 1 is $w_0 g_0$, and pairing with
 611 it is the weighted inner product, $\int_0^\infty w_0 g_0 (T_k g) dy = \langle g_0, T_k g \rangle_{w_0}$ with $\langle f, g \rangle_{w_0} = \int_0^\infty f g w_0 dy$.
 612 The coefficients are therefore $\lambda^{[1]} = \langle g_0, T_1 g_0 \rangle_{w_0} / \langle g_0, g_0 \rangle_{w_0}$ and $\lambda^{[2]} = (\langle g_0, T_2 g_0 \rangle_{w_0} + \langle g_0, (T_1 -$
 613 $\lambda^{[1]})g_1 \rangle_{w_0}) / \langle g_0, g_0 \rangle_{w_0}$, where g_1 solves $(T_0 - 1)g_1 = -(T_1 - \lambda^{[1]})g_0$ with $\langle g_0, g_1 \rangle_{w_0} = 0$; the
 614 symmetric case identities that move T_1 to the other side of an inner product are not available, but they
 615 are not needed below. First, $\lambda^{[0]} = 1$. Second, $T_1 g_0 = -\frac{n-1}{3}(y+y) = -\frac{2}{3}(n-1)g_0$ exactly, so
 616 $\lambda^{[1]} = -\frac{2}{3}(n-1)$ and the first order correction to the eigenfunction vanishes, because the equation
 617 $(T_0 - 1)g_1 = -(T_1 - \lambda^{[1]})g_0 = 0$ has only the solution $g_1 \in \text{span}\{g_0\}$. Third, with $g_1 = 0$ the
 618 second order coefficient reduces to

$$\lambda^{[2]} = \frac{\langle g_0, T_2 g_0 \rangle_{w_0}}{\langle g_0, g_0 \rangle_{w_0}} = \frac{4(n-1)}{45} \frac{\int_0^\infty y^{n+3} e^{-y^2/2} dy}{\int_0^\infty y^{n+1} e^{-y^2/2} dy} = \frac{4(n-1)(n+2)}{45},$$

619 using $T_2 g_0 = (n-1)(\frac{1}{45} + \frac{1}{15})y^3 = \frac{4(n-1)}{45}y^3$ and $\int_0^\infty y^m e^{-y^2/2} dy = 2^{(m-1)/2} \Gamma(\frac{m+1}{2})$, so
 620 that the moment ratio equals $2\Gamma(\frac{n+4}{2})/\Gamma(\frac{n+2}{2}) = n+2$. The radial statement follows from
 621 Proposition 3.3(iii); directly, the radial sector has $T_1^{(0)} = -\frac{n-1}{3}y\partial_y$ acting on the radial ground state
 622 $y^2 - n$, which is not an eigenfunction of $T_1^{(0)}$, and the intertwining identity is the shorter route.

623 **Cross-check by the ground state transform.** The same coefficients follow from a symmetric
 624 problem in which no left eigenvector is needed. The unitary map $f \mapsto \rho_c^{1/2} f$ from $L^2(\rho_c ds)$ to
 625 $L^2(ds)$ carries $-\mathcal{L}_c^{(1)}$ to the Schrödinger operator $H = -\partial_s^2 + a^2/4 + a'/2 + (n-1)\sinh^{-2}s$. In
 626 the variable y , $H = cH_0 + H_1 + c^{-1}H_2 + O(c^{-2})$ with

$$H_0 = -\partial_y^2 + \frac{y^2}{4} - \frac{n}{2} + \frac{n^2-1}{4y^2}, \quad H_1 = -\frac{n-1}{6}(y^2 - (n-2)), \quad H_2 = (n-1)\left(\frac{y^4}{90} + \frac{(n+1)y^2}{60}\right),$$

627 all symmetric on $L^2(dy)$. H_0 is the radial harmonic oscillator with eigenvalues $1+2k$ and ground
 628 state $u_0 = y^{(n+1)/2}e^{-y^2/4}$, under which y^2 and y^4 have means $n+2$ and $(n+2)(n+4)$. Here H_1
 629 does not have u_0 as an eigenfunction, so the first order correction to the eigenfunction is nonzero and
 630 the second order coefficient carries a sum over states. Since $(y^2 - (n+2))u_0$ is proportional to the
 631 first excited state u_1 , that sum has a single term with energy denominator 2, and

$$\lambda^{[1]} = -\frac{(n-1)(n+2)}{6} + \frac{(n-1)(n-2)}{6} = -\frac{2}{3}(n-1),$$

$$\lambda^{[2]} = \frac{(n-1)(n+2)(n+4)}{90} + \frac{(n-1)(n+1)(n+2)}{60} - \frac{(n-1)^2(n+2)}{36} = \frac{4(n-1)(n+2)}{45},$$

632 using $\text{Var}_{u_0}(y^2) = 2(n+2)$ in the last term. The two routes distribute the contributions differently
 633 and agree; the supplied script `check_rs.py` carries out the second route symbolically.

634 The expansion is formal in two respects: the series for $\coth$ and $\sinh^{-2}$ converge only for $|y| <$
 635 $\pi\sqrt{c}$, and T_1, T_2 are unbounded perturbations of T_0 . Both are standard features of the harmonic
 636 approximation for low lying eigenvalues, for which [22] gives the asymptotic expansion in the
 637 Schrödinger setting, but we have not carried the argument through for the present operator.

638 C Experimental details

639 The core experiment script uses NumPy in double precision on a single CPU, with Matplotlib for
 640 plotting and SciPy, SymPy and mpmath in the auxiliary verification scripts. The reported experiment
 641 scripts run in under an hour at the stated CPU scale. Seeds are fixed in the supplementary code.

642 C.1 Relaxation rates (Section 5.1) and numerical spectral gaps (Table 1)

643 **Integrator.** We integrate the forward SDE on $\mathbb{H}^2$ in exact polar coordinates: $dr = (\coth r - cr) dt +$
 644 $\sqrt{2} dW_r$ with reflection at $r = 0$ (Bessel-type), and $d\varphi = (\sqrt{2}/\sinh r) dW_\varphi$. We use $N = 20,000$
 645 particles, $h = 0.005$ and $T = 6$, clipping the drift at ± 50 only in the reflection region. Initially,
 646 $r \sim |\mathcal{N}(3, 0.2^2)|$ and $\varphi \sim \mathcal{N}(0, (0.2/\sinh 3)^2)$, which gives a small cloud at radius 3 and angle 0.

647 **Observable and fits.** We estimate $R(t) = |\mathbb{E} e^{i\varphi_t}|$ over particles and take rates as least-squares
 648 slopes of $\log R$ on the common window $0.05 \leq R \leq 0.7$, with $t > 0.05$. The same window is
 649 used for every c . For $c \in \{0.05, 0.1\}$ the trajectory does not enter this window within the horizon;
 650 the plotted open triangles therefore use slopes over $[T/2, T]$ and are only transient limited upper
 651 estimates, as stated in the caption. For reference, the small c envelope $\sqrt{c}e^{-1/(2c)}$ predicts a time
 652 constant of order 5×10^2 at $c = 0.1$, two orders beyond the simulated horizon.

653 **Cross-checks.** A replicate initialised at the radial equilibrium $r_*(c)$, the root of $\coth r = cr$,
 654 reproduces the fitted rates within about 20% whenever the fit window applies (the largest deviation,
 655 0.044 against 0.037, is at $c = 0.25$). We ran an independent integrator in full hyperboloid coordinates.
 656 It uses geodesic Euler–Maruyama with projection after every step. At $c = 1$, starting from the same
 657 initial law, it gives $R(T) = 0.053$ compared with 0.049 from the polar integrator. Both values are
 658 near the Monte Carlo floor $N^{-1/2} \approx 0.007$ for the estimator of $|\mathbb{E} e^{i\varphi}|$ near zero.

659 **Numerical spectral gaps.** The values in Table 1 are the lowest eigenvalues of the sector forms
 660 $\mathcal{E}_\ell$ of (1), obtained by a linear finite element discretisation on a uniform grid over $[0, R]$ with
 661 $R = \max\{9, 1/c + 9/\sqrt{c}\}$ and 8,000 elements, with two point Gauss quadrature per element and
 662 the weight ρ_c evaluated in log space. The radial sector uses natural boundary conditions and the
 663 second eigenvalue is reported; the angular sectors use a Dirichlet condition at $r = 0$ and the first
 664 eigenvalue is reported. Doubling the number of elements changes the values by less than 10^{-6}
 665 relative. The radial variance $\text{Var}_{\mu_c}(r)$ in the bound of Corollary 3.4 is computed by quadrature of ρ_c
 666 on the same interval. The exponent fits quoted in Section 3.4 use the same discretisation with 12,000
 667 elements and $R = \max\{9, (n-1)/c + 10/\sqrt{c}\}$; they are least squares slopes of $\log \lambda_1(c)$ against
 668 $1/c$ over $c \in \{0.05, \dots, 0.125\}$ for $n = 2$ (with $\frac{1}{2} \log c$ subtracted), $c \in \{0.10, \dots, 0.22\}$ for $n = 3$,
 669 $c \in \{0.18, \dots, 0.35\}$ for $n = 4$ and $c \in \{0.26, \dots, 0.45\}$ for $n = 5$. The verification scripts require
 670 NumPy 2.

671 **Higher dimensions.** Table 6 repeats the computation on $\mathbb{H}^3$, $\mathbb{H}^4$ and $\mathbb{H}^5$ with 12,000 elements and
 672 $R = \max\{9, (n-1)/c + 10/\sqrt{c}\}$. The residual $\lambda^{(0)} - \lambda^{(1)} - c$ of Proposition 3.3 is below 10^{-5} in
 673 absolute value in every entry (below 1.5×10^{-4} on $\mathbb{H}^2$ at $c = 16$), and it grows with c because the
 674 eigenfunctions sharpen at scale $c^{-1/2}$ on a fixed grid. Doubling the number of elements changes the
 675 entries by less than 10^{-7} relative, except for the two entries below 10^{-5} , which change by less than
 676 3×10^{-4} relative. The large c values quoted in Appendix B.7 use $R = \min\{9, 1 + 12/\sqrt{c}\}$ with
 677 16,000 elements, and the limits of the scaled remainder are Richardson extrapolations from $c = 128$
 678 and $c = 256$; doubling the number of elements changes the eigenvalues by less than 3×10^{-9} .

679 **Relaxation rates in higher dimension.** For Table 7 we integrate the polar form of the forward
 680 process on $\mathbb{H}^n$ by Euler–Maruyama in the radial coordinate, reflected at $r = 0$, and by a geodesic
 681 random walk on S^{n-1} with step variance $2h \sinh^{-2} r$ per tangent coordinate, which is exact for
 682 $n = 2$ and first order accurate in general. Step sizes are $h = 0.002$ for $c \leq 2$ and $h = 0.001$
 683 for $c \geq 4$, with 200,000 particles and four independent replicates per entry; the table reports the
 684 replicate mean and standard deviation. The angular runs start from the radial equilibrium law with the
 685 direction concentrated near e_1 , and the rate is the least squares slope of $\log \mathbb{E} \theta_1(X_t)$ on the window
 686 $0.05 \leq \mathbb{E} \theta_1 \leq 0.2$. The radial runs start from the radial equilibrium law shifted outwards by 0.3,
 687 and the rate is the slope of $\log |\mathbb{E} r_t - \mathbb{E}_{\mu_c} r|$ on the window $[0.02, 0.15]$; starting all particles at one
 688 radius instead excites transient radial modes of the opposite sign and biases this slope low by up to
 689 5% at large c . Entries marked n/a are cases in which the angular decay does not reach the fitting

Table 6: Spectral gap of $-\mathcal{L}_c$ on $\mathbb{H}^n$ for $n = 3, 4, 5$ from the sector problems (1), with the bounds of Section 3. The residual row tests Proposition 3.3(i). Entries marked n/a are values of c at which a bound is vacuous.

c	0.25	0.5	1	2	4	8	16
$\mathbb{H}^3$ ($\beta(3) = 2$, threshold $c = 2$)							
$\lambda_1(c)$, numerical	7.91×10^{-4}	0.0346	0.268	1.032	2.869	6.773	14.72
residual $\lambda^{(0)} - \lambda^{(1)} - c$	-4×10^{-11}	5×10^{-10}	1×10^{-8}	6×10^{-8}	4×10^{-7}	2×10^{-6}	8×10^{-6}
upper bound $\text{Var}_{\mu_c}(r)^{-1} - c$	8.39×10^{-4}	0.0403	0.330	1.282	3.532	8.263	17.86
ratio $\lambda_1/(\text{Var}^{-1} - c)$	0.943	0.859	0.812	0.805	0.812	0.820	0.824
lower bound $c - (n - 1)$	n/a	n/a	n/a	0	2	6	14
large c expansion $c - \frac{2}{3}(n - 1)$	n/a	n/a	n/a	0.667	2.667	6.667	14.67
$\mathbb{H}^4$ ($\beta(4) = 4$, threshold $c = 3$)							
$\lambda_1(c)$, numerical	1.23×10^{-6}	2.74×10^{-3}	0.101	0.681	2.375	6.195	14.10
residual $\lambda^{(0)} - \lambda^{(1)} - c$	2×10^{-12}	-5×10^{-11}	2×10^{-9}	3×10^{-8}	2×10^{-7}	1×10^{-6}	5×10^{-6}
upper bound $\text{Var}_{\mu_c}(r)^{-1} - c$	1.23×10^{-6}	2.86×10^{-3}	0.114	0.800	2.786	7.205	16.30
ratio $\lambda_1/(\text{Var}^{-1} - c)$	0.996	0.959	0.882	0.851	0.853	0.860	0.865
lower bound $c - (n - 1)$	n/a	n/a	n/a	n/a	1	5	13
large c expansion $c - \frac{2}{3}(n - 1)$	n/a	n/a	n/a	0	2	6	14
$\mathbb{H}^5$ ($\beta(5) = 6$, threshold $c = 4$)							
$\lambda_1(c)$, numerical	6.01×10^{-10}	9.38×10^{-5}	0.0284	0.419	1.933	5.642	13.49
residual $\lambda^{(0)} - \lambda^{(1)} - c$	2×10^{-12}	6×10^{-12}	1×10^{-10}	1×10^{-8}	1×10^{-7}	8×10^{-7}	4×10^{-6}
upper bound $\text{Var}_{\mu_c}(r)^{-1} - c$	6.04×10^{-10}	9.43×10^{-5}	0.0302	0.472	2.199	6.378	15.15
ratio $\lambda_1/(\text{Var}^{-1} - c)$	0.996	0.995	0.942	0.887	0.879	0.885	0.890
lower bound $c - (n - 1)$	n/a	n/a	n/a	n/a	0	4	12
large c expansion $c - \frac{2}{3}(n - 1)$	n/a	n/a	n/a	n/a	1.333	5.333	13.33

690 window within the horizon $T = 6$. The $\mathbb{H}^2$ angular rates are re-estimated under this protocol as a
691 control and agree with Table 1 to within 3%. On $\mathbb{H}^2$ the radial rates fitted to $\mathbb{E} r_t$ are low by 1% to
692 6%, and Table 8 locates the cause. Any observable that is an exact eigenfunction of $-\mathcal{L}_c$ decays as
693 a single exponential for every initial law, $\mathbb{E} f(X_t) = e^{-\lambda t} \mathbb{E} f(X_0)$, so its fitted slope contains no
694 transient and measures only the integrator's bias and the Monte Carlo noise. We take g_0 and g_1 from
695 the finite element eigenvectors of the radial and first harmonic sectors, record $\mathbb{E} g_0(r_t)$ and $\mathbb{E} g_1(r_t)\theta_1$
696 alongside $\mathbb{E} r_t$ and $\mathbb{E} \theta_1$ on the same trajectories, and fit the eigen-observables on the window between
697 60% and 5% of their initial value. Two integrators are compared: the polar scheme above at three
698 step sizes, and a geodesic Euler scheme on the hyperboloid, $x \leftarrow \exp_x(-cr \nabla r h + \sqrt{2h} \xi)$ with ξ a
699 standard Gaussian in $T_x \mathbb{H}^2$, which has no singular drift at the origin. On $\mathbb{H}^2$ the eigen-observable
700 ratios sit within 2% of 1 in 21 of the 22 entries and within 3.5% in the remaining one, with no trend
701 in the step size and no difference between the integrators, while the raw $\mathbb{E} r_t$ ratios on the same
702 trajectories fall to 0.95 at $c = 2$. The same protocol on $\mathbb{H}^3$ and $\mathbb{H}^4$ gives ratios within 1.5% in all 16
703 entries, while the raw $\mathbb{E} r_t$ ratio on the same trajectories is 0.95 at $c = 8$. The deviation in Table 7 is
704 therefore a property of fitting a multi-exponential signal on a fixed window, and the integrators are
705 accurate at the percent level down to the origin. With the eigen-observables, the difference of the
706 radial and angular rates divided by c lies between 0.97 and 1.02 in all 19 cases, against the value 1 of
707 Proposition 3.3(iii).

708 **Conditioning protocol.** For Table 9, points at radius r are constructed directly on the hyperboloid
709 in each precision at 512 angles. We maximise the constraint residual $|\langle x, x \rangle_L + 1|$ over those angles.
710 The drift figures in Section 4.3 are maximal constraint residuals over 400 geodesic Euler–Maruyama
711 steps at $c = 2$ and $h = 0.01$. These runs start from μ_2 and do not use reprojection.

712 C.2 Hierarchical generation (Section 5.2)

713 **Data.** Groups $k \in \{0, 1, 2, 3\}$ at angles $k\pi/2$ with weights (0.4, 0.3, 0.2, 0.1); two leaves per group
714 at angular offsets ± 0.175 ; leaf centres $\exp_o(3.5(\cos \vartheta, \sin \vartheta, 0))$; samples $\exp_{\text{centre}}(0.15 \xi)$ with ξ a
715 standard tangent Gaussian. 8,192 training and 4,096 test points.

Table 7: Simulated relaxation rates on $\mathbb{H}^n$ against the sector gaps of Proposition 3.3: the angular observable $\mathbb{E} \theta_1(X_t)$ should decay at $\lambda_1(c)$ and the radial observable $\mathbb{E} r_t - \mathbb{E}_{\mu_c} r$ at $\lambda_1(c) + c$, so the last column should equal 1. Rates are means $\pm$ standard deviations over four replicates of 200,000 particles.

n	c	$\lambda_1(c)$ (FEM)	angular rate	ratio	$\lambda_1(c) + c$	radial rate	(radial $-$ angular)/ c
2	0.5	0.1762	n/a	n/a	0.6762	0.663 ± 0.019	n/a
2	1	0.5645	0.567 ± 0.008	1.005	1.5645	1.488 ± 0.038	0.921
2	2	1.4736	1.471 ± 0.056	0.998	3.4736	3.266 ± 0.071	0.897
2	4	3.4118	3.472 ± 0.075	1.018	7.4118	7.334 ± 0.121	0.965
3	1	0.2683	0.271 ± 0.015	1.008	1.2683	1.274 ± 0.051	1.003
3	2	1.0324	1.036 ± 0.015	1.004	3.0324	3.026 ± 0.013	0.995
3	4	2.8691	2.877 ± 0.056	1.003	6.8691	6.836 ± 0.142	0.990
3	8	6.7729	6.869 ± 0.084	1.014	14.7729	14.467 ± 0.211	0.950
4	1	0.1009	n/a	n/a	1.1009	1.122 ± 0.022	n/a
4	2	0.6811	0.684 ± 0.005	1.004	2.6811	2.654 ± 0.076	0.985
4	4	2.3751	2.402 ± 0.058	1.011	6.3751	6.312 ± 0.211	0.978
4	8	6.1949	6.286 ± 0.082	1.015	14.1949	13.918 ± 0.237	0.954

Table 8: Exact eigenfunction observables. Ratios of fitted rate to finite element eigenvalue for the eigen-observables $g_0(r_t)$ and $g_1(r_t)\theta_1$ (mean $\pm$ standard deviation over three replicates of 60,000 particles) next to the raw observables of Table 7 on the same trajectories, for two integrators and several step sizes h on $\mathbb{H}^2$ and for the polar integrator on $\mathbb{H}^3$ and $\mathbb{H}^4$. The last column is the eigen-observable test of Proposition 3.3(iii), (radial rate $-$ angular rate)/ c , which should equal 1. Entries marked n/a did not reach the fitting window within the horizon.

n	c	integrator	h	radial, eigen g_0	radial, raw $\mathbb{E} r_t$	angular, eigen $g_1\theta_1$	angular, raw $\mathbb{E} \theta_1$	(radial $-$ angular)/ c
2	0.5	polar	0.002	1.006 ± 0.030	0.971	1.010 ± 0.017	n/a	1.005
2	0.5	polar	0.001	0.982 ± 0.017	0.967	1.005 ± 0.033	n/a	0.974
2	0.5	hyperboloid	0.002	0.995 ± 0.029	0.985	0.988 ± 0.014	n/a	0.997
2	0.5	hyperboloid	0.001	0.981 ± 0.033	0.981	0.996 ± 0.014	n/a	0.976
2	1	polar	0.002	0.991 ± 0.003	0.996	1.020 ± 0.019	1.036	0.975
2	1	polar	0.001	1.019 ± 0.007	1.022	1.035 ± 0.009	1.060	1.010
2	1	hyperboloid	0.002	0.998 ± 0.012	0.981	0.993 ± 0.023	0.996	1.001
2	1	hyperboloid	0.001	0.988 ± 0.005	1.004	0.990 ± 0.055	0.985	0.987
2	2	polar	0.002	0.999 ± 0.010	0.948	1.012 ± 0.042	1.014	0.989
2	2	polar	0.001	0.998 ± 0.005	0.966	0.994 ± 0.002	0.992	1.001
2	2	hyperboloid	0.002	1.007 ± 0.005	0.985	0.995 ± 0.032	1.011	1.016
3	1	polar	0.002	1.003 ± 0.012	0.984	1.015 ± 0.008	1.103	1.000
3	2	polar	0.002	1.000 ± 0.009	0.989	1.002 ± 0.016	1.002	0.999
3	4	polar	0.001	0.998 ± 0.003	0.977	1.015 ± 0.013	1.014	0.986
3	8	polar	0.001	1.003 ± 0.004	0.947	0.999 ± 0.019	0.995	1.006
4	1	polar	0.002	1.008 ± 0.012	1.036	1.008 ± 0.018	n/a	1.008
4	2	polar	0.002	0.991 ± 0.009	0.998	0.998 ± 0.010	0.987	0.989
4	4	polar	0.001	1.001 ± 0.005	0.983	1.007 ± 0.036	1.009	0.997
4	8	polar	0.001	1.003 ± 0.002	0.953	1.005 ± 0.008	0.996	1.001

Networks and features. All models use a tanh multilayer perceptron with two hidden layers of 128 units, trained with Adam at learning rate 10^{-3} , reduced to 3×10^{-4} after 20k of 30k iterations, and batch size 512. The time embedding contains t/T together with $\sin(2\pi kt/T)$ and $\cos(2\pi kt/T)$ for $k = 1, \dots, 4$, giving 9 features. The intrinsic model takes $\min(r, 12)$, angular Fourier features $\cos k\varphi, \sin k\varphi$ for $k \in \{1, 2, 4, 8\}$ and the time embedding, and outputs the frame coefficients (a_θ, b_θ) from Section 4.2. The tangent baseline takes $y \in \mathbb{R}^2, |y|$, the same Fourier features of $y/|y|$ and the same time embedding, ensuring that both models receive the same feature family.

Intrinsic training. We simulate a bank of forward trajectories once from the training set, with $K = 400$ steps, $h = 0.01$ and $T = 4$. For each batch element, we draw a start index k and a span $m \in \{1, 2, 4, 8\}$ uniformly, then use the target score $\hat{s} = \log_{x_{k+m}}(\Phi_{mh}(x_k))/(2mh)$ at x_{k+m} . Here

Table 9: Hyperboloid constraint error $\max |\langle x, x \rangle_L + 1|$ for points constructed at radius r (512 angles). The representation fails near $r \approx \frac{1}{2} \log(2/\varepsilon_{\text{mach}})$: about 8 in single precision and 18 in double.

r	2	5	8	12	16	20	24
fp32	1.8×10^{-6}	9.2×10^{-4}	0.36	1.3×10^3	3.8×10^6	8.6×10^9	2.4×10^{13}
fp64	5.3×10^{-15}	3.6×10^{-12}	4.7×10^{-10}	1.9×10^{-6}	7.8×10^{-3}	1.7×10^1	6.6×10^4

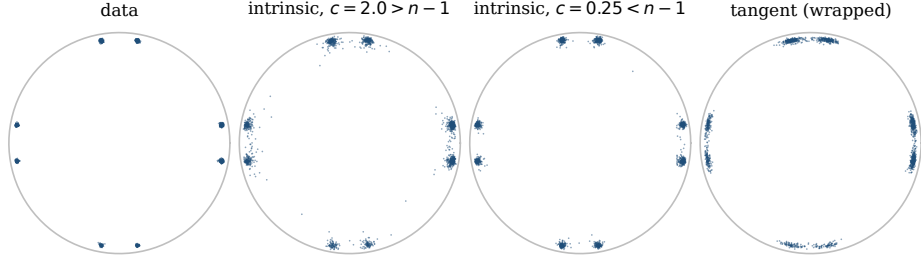


Figure 2: Data and generated samples in the Poincaré disk. Above threshold ($c = 2.0$) the intrinsic model resolves all eight leaves with the correct group weights; below threshold ($c = 0.25$) the same architecture, budget and estimator smear and bridge adjacent leaves, since the forward process never destroyed the angular structure that the reverse process must reallocate. The tangent baseline resolves the leaves with the tangential elongation of the $\log_o$ chart.

726 Φ_τ is the geodesic flow of the drift alone, computed by m integrator steps, and the residual target
727 is $u^* = \hat{s} + cr e_r$. The loss is the batch mean of $2\tau |u_\theta - u^*|_g^2$ with $\tau = mh$; the factor 2τ scales
728 the target to unit variance across spans, so that no span dominates the objective. Approximating the
729 conditional mean introduces $O(mh)$ bias, whereas increasing m reduces variance by a factor of order
730 $\sqrt{m}$; the largest span gives $mh = 0.08$.

731 **Baseline training and sampling.** The baseline is a variance preserving OU diffusion on $y =$
732 $\log_o x \in \mathbb{R}^2$ with exact Gaussian conditionals, trained by exact denoising score matching using ε
733 prediction at the same budget. Sampling uses 400 reverse Euler–Maruyama steps in the tangent plane
734 followed by $\exp_o$, whereas the intrinsic sampler uses 400 reverse geodesic Euler–Maruyama steps of
735 (2) with the anchored score. Its initial radius comes from numerical inverse transform of the exact
736 radial density of μ_c ; the released code evaluates the CDF on a fine grid over $[0, 14]$ and interpolates it.
737 The sampler caps the geodesic step norm at 0.5 and the drift norm at 50, with a numerical wall at
738 $r = 30$. In the final models the drift cap never engages, the wall is never reached, and the step cap
739 clips 0.21% of steps, close to the Gaussian tail probability $\mathbb{P}(\sqrt{2h}|\xi|_2 > 0.5) \approx e^{-6.25}$.

740 **Metrics.** MMD² is the unbiased U-statistic with an RBF kernel on squared hyperbolic distance,
741 with bandwidth set by the median heuristic on the test data. The estimator can be slightly negative,
742 and we report those values as computed. Leaf total variation assigns each generated point to its
743 nearest true leaf centre in hyperbolic distance and compares the resulting histogram with the true
744 leaf weights. Divergence is the fraction of samples with $r > 6$; the data lie at $r \approx 3.5$. Reported
745 uncertainties are means and standard deviations over three fixed sampling seeds with fixed trained
746 networks. The forward diagnostics use the angle φ of each trajectory endpoint on the hyperboloid:
747 $R_4(T) = |\mathbb{E} e^{4i\varphi T}|$ over the 8,192 training trajectories, and the prior mismatch is the unbiased MMD²
748 between the first 4,096 endpoints and an independent sample of μ_c of the same size, with the same
749 kernel as above.

750 D Full ablation and diagnostic results

751 **Parameterisation ladder.** The reduced budget ablations use the same data, architecture width,
752 sampler and sampling seed. The unanchored extrinsic model was trained for 4k iterations, whereas

Table 10: Ladder of score parameterisations at $c = 2.0$ using a single sampling seed. The unanchored extrinsic row was trained for 4k iterations; the anchored extrinsic and anchored frame rows use 6k iterations, all with single span targets. The final row uses 30k iterations with Fourier features and multi-span targets. Divergence is the fraction of samples with $r > 6$.

parameterisation	MMD ²	leaf TV	diverged
extrinsic, unanchored	0.0773	0.412	28.7%
extrinsic, anchored	0.0452	0.186	27.1%
frame, anchored	0.0168	0.090	0.0%
frame, anchored + Fourier + multi-span (30k)	0.0009	0.055	0.0%

the anchored extrinsic and frame variants use 6k; only those two are budget matched. Extrinsic parameterisations output $w \in \mathbb{R}^3$ and project it with P_x ; the anchored variant adds $-\nabla W_c$ and projects only the residual. In diverging trajectories the radius grows as network inputs leave the training support, after which the $\Theta(e^r)$ amplification of Lemma 4.1 becomes dominant. The frame variant instead predicts two coefficients multiplying (e_r, e_φ) .

Forward diagnostics. At $T = 4$, the $c = 2.0$ process has $R_4(T) = 0.010$ and $\mathbb{E}r(T) = 0.975$, consistent with the stationary mean radius 0.96 of μ_2 . At $c = 0.25$, $R_4(T) = 0.380$ compared with the data value 0.765, while the mean radius $\mathbb{E}r(T) = 3.91$ has moved from the initial 3.5 most of the way to the equilibrium mean 4.19. The forward process has therefore relaxed radially far more than angularly, the pattern described by Theorem 3.5 and quantified by Proposition 3.3: at $c = 0.25$ the radial rate $\lambda^{(0)} = 0.29$ gives a relaxation time close to the horizon, whereas the angular gap 0.038 gives a time scale of about 26. The prior mismatch is -1×10^{-5} at $c = 2.0$, which is zero within estimator noise, and 1.59×10^{-2} at $c = 0.25$. The diagnostic uses only the forward process; if the mismatch remains appreciable at the chosen horizon, the reverse sampler is initialised from a law that the forward process has not reached.

Repeatability. Across the three fixed sampling seeds, the below threshold model is worse in every run summarised by Table 2. For both MMD² and leaf TV, the difference between $c = 0.25$ and $c = 2.0$ is about seven times the combined reported sampling standard deviation, while the $c = 2.0$ model and the tangent baseline remain close on MMD² at the scale of this sampling variation. These uncertainties come from sampling seeds with fixed trained networks and are not full training uncertainties or formal significance tests.

Code availability

All code, result files and verification scripts are available anonymously at <https://anonymous.4open.science/status/2026-STODY-NeurIPS-F6C4>.

Disclosure of AI assistance

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